TM Forum Exploratory Report

ODA Functional Architecture Exploratory Report

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1. SCOPE OF THIS DOCUMENT

1.1. Purpose

The Purpose of this document is to explore and introduce structured concepts which are implementation-neutral using simplified views of business information objects interacting in an Information Systems environment.

The document is intended for technical stakeholders that explore new functions, new capabilities, end-to-end business information objects, and related technologies to realize the concepts and principles defined in the Open Digital Architecture best practices and standard. System architects, solution architects, product designers, systems integration specialists et al. who are exploring or advancing the functional requirements of Digital Organizations can firstly use this document to landscape their insights and approach.

The functional architecture, TMF GB1022, enforces the concepts and principles established in GB998 Open Digital Architecture (ODA) Concepts & Principles, which outline characteristics that are universal to leading digital businesses. These characteristics include decoupling, componentization and exposure of capabilities. This document is used to share concepts that advance use cases of ODA and to propose new best practices and standards that can be adopted into the Standard - GB1022 ODA Functional Architecture Guidebook.

By continuing to capture and rationalize ODA best practices, the business requirements from service providers and concerns of suppliers and integrators are altogether addressed. Through this document, the industry is able to identify and propose new best practices as the dynamics of business and operations change.

1.2. Scope

The scope of the Introductory Guide for ODA Functional Architecture is to:

- Advance taxonomy of Business and Infrastructure Functions
- Specify the functionality of those Functions, advance their incorporation or mapping to Frameworx.
- Explore scenarios for using the Functions based on implementation that is technology neutral aka Features/ Feature Groups (SID information Framework Definition)

This will enable:

- Digital Enterprises/Organizations to define new business requirements based on an unambiguous set of Functions - linked to Frameworx - For consistency in expressing requirements through reuse of function specifications.
- Technology ecosystems to interpret business requirements based on one language.
- The creation of reusable ODA Component specifications from groupings of these functions
- Facilitate iterative and incremental design and implementation by developers.

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1.3. Benefits of ODA - Functional Architecture

- Enforces separation of concerns to enable technical and commercial evolution.
- Easy to realize cross channel capabilities by reusing Systems of Record (SoR) for any and all Systems of Engagement (SoE).
- Eliminates data replication between Functions that process the same data.
- Enabled Business model innovation (Supports different Actor/Roles such as Retailer, Biller, … beyond the pipeline/linear business model)
- Enables platformization of Functions through standardized definition of API (TMF) at the border of Function blocks in order to expose functionalities and services in a unified and harmonized way.

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2. CONVENTIONS AND TERMINOLOGY

2.1. Conventions

All conventions follow industry standards. Where a convention is unique, it will be adequately described based on the topic and domain.

2.2. Concepts

Reference GB998 Open Digital Architecture (ODA) Concepts & Principles v3.0.0 (ODA-645)

2.3. Terminology

Reference TMF071 ODA Terminology R19.0.1

2.4. Definitions

Reference TMF071 ODA Terminology R19.0.1

2.5. Abbreviations

Reference TMF071 ODA Terminology R19.0.1

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3. INTRODUCTION TO ODA

Reference GB1022 ODA Functional Architecture Guidebook

3.1. The Functional Architecture

The Functional Architecture has been modeled to represent enterprise functions and corresponding interactions that go beyond the traditional pipe-line business models. Each of the ODA Function Blocks is an enterprise level group of functions with a collection of coherent processes continuously performed within the enterprise to support its mission.

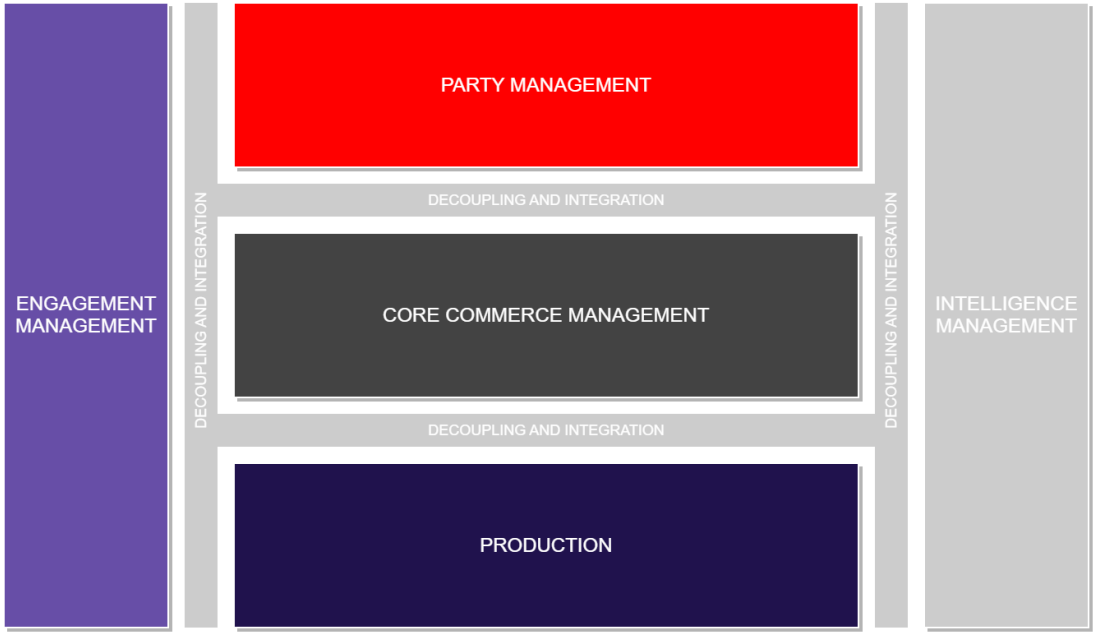

Figure 3.1. ***ODA Functional Architecture - High Level view (Level 0)***

3.2. Systems of Separation

The functional view of ODA can be summarized in a system view based on the architectural elements that deliver the system's functionality. The view in Figure 3-2 below groups the ODA function blocks into a system's structure-including the key functional elements (Party Management, Core Commerce Management, and Production), their responsibilities, which can be used to summarize their roles, interfaces they expose, and the interactions between them. Taken together, this shows how the responsibility of the systems and the functions required of them.

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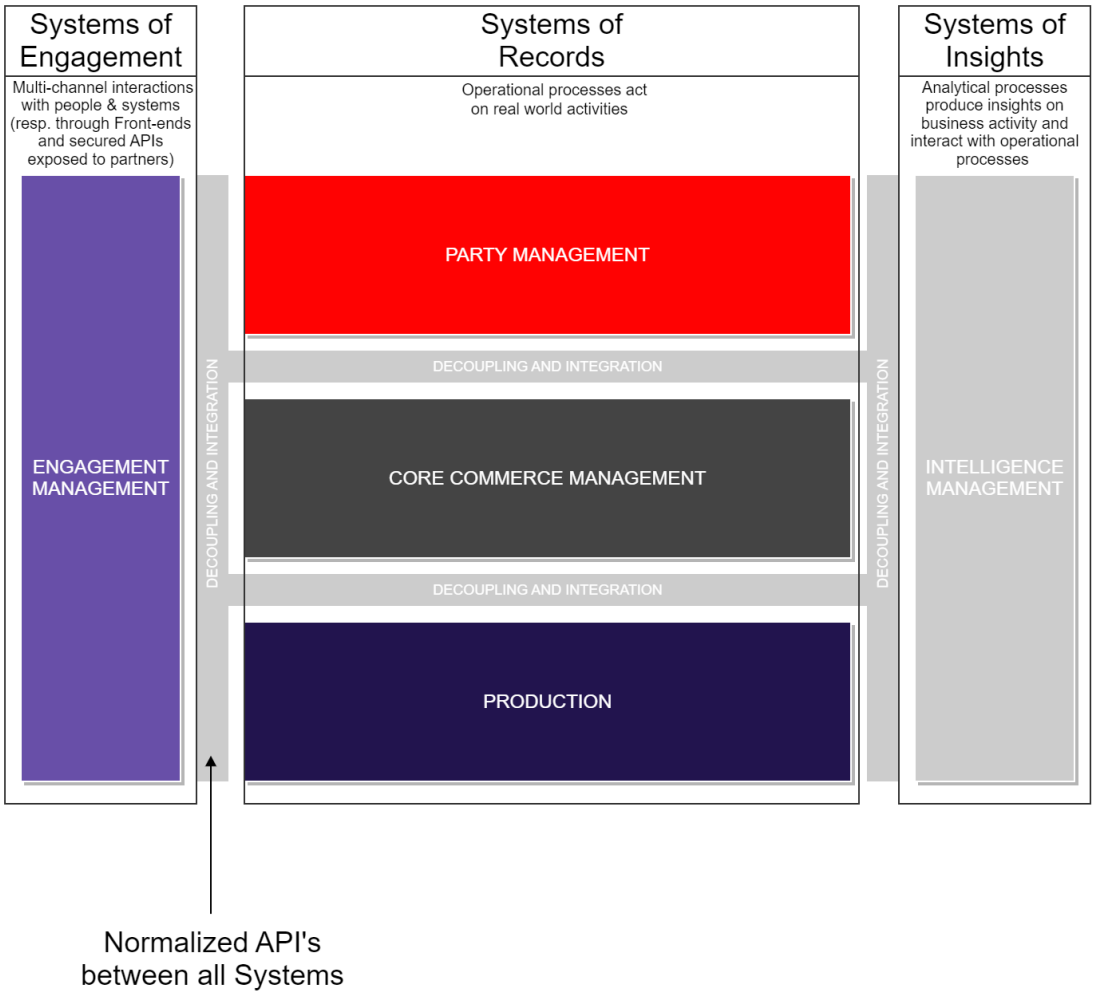

Figure 3.2. ***Grouping of systems based on the Functional Architecture***

Rationale of systems separation

Each of above defined system types has its own lifecycle :

- Engagement Management deals with user interactions that are powered by fast moving technologies, it is committed to evolve (e.g., in a quick & agile mode) in order to catch the trends on the fields of customer experience, ergonomics, social networking. These trends apply to all industries not only telecommunications, therefore Service Providers have no control over them and must adapt quickly.
- Conversely, Systems of Records hold assets (processes, objects and data) that need stability over time, while maintaining a strong ability to interwork with fast moving Engagement systems. Records is therefore responsible for in variants, stability over time, and integrity of data. Hence, Records goes through a slower lifecycle.
- Intelligence Management, although it is Back-end systems by nature, is as fast-evolving as Engagement Management, due in particular to technology leaps such as Big & Fast Data that enables functionality previously out of reach of companies (too expensive or needing extreme processing capabilities). Therefore, it makes sense to segregate them from the other two.

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4. BUSINESS USE CASES

This set of Business oriented use cases illustrates how ODA can solve current business issues shared by many CSPs.

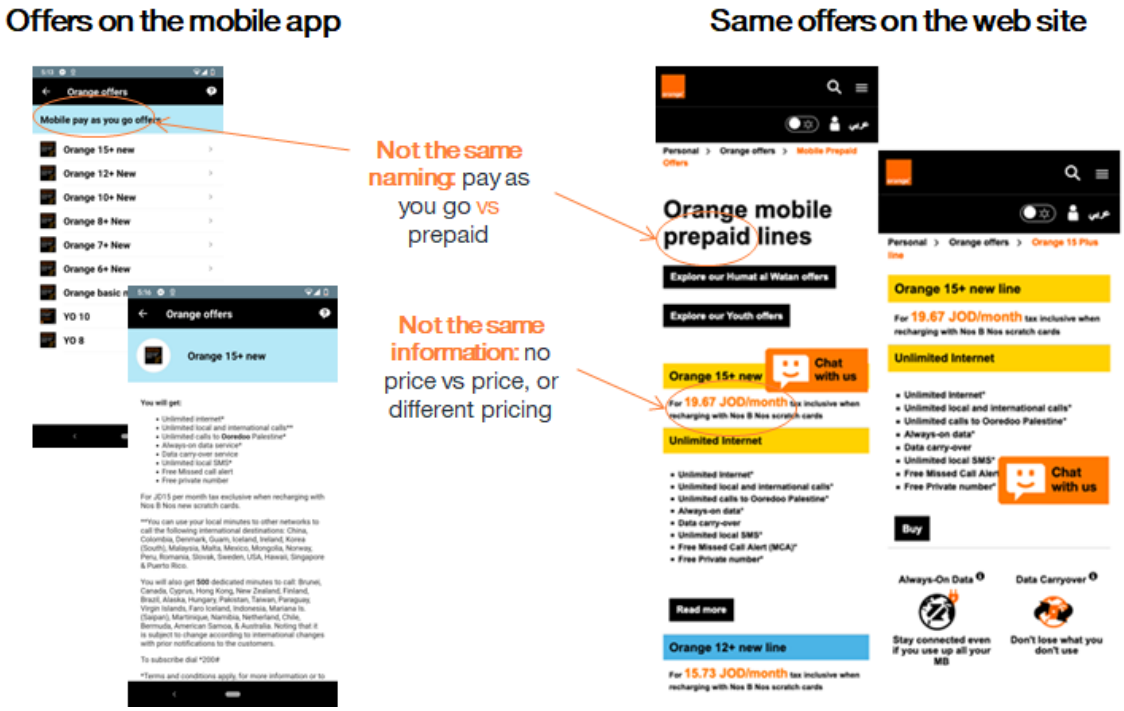
4.1. Multi-Channel / Omni-Channel User Story

Multi and omni channel capabilities and presentation consistency is one of the major issues of digitalization for CSPs. Historically each channel has been managed by dedicated applications that generally do not permit to address the issue. We will see through these use cases how ODA decoupling between Engagement Management, in charge of GUI kinematics, and process layers will help to solve it.

4.1.1. Multi-channel Catalogue Management Use Case

Usually, with CSPs, when queries are performed on one channel interface, the results differ, most often, with other channels. This can, and most often poses a challenge to customers, partners and service delivery personnel (like call center agents, and back office customer support staff).

While presentation of content from, say catalogs, can be optimized per channel, device and/or customer type/segments, the need to maintain consistency across all these channels for objects like Offer names, and/or attributes are key to drive unified experience and customer engagement consistency. Below is a reference of some of the issues.



The reason of these differences, or even inconsistencies, between the information provided on each channel is that CSPs often have implemented applications or components dedicated to one channel and covering GUI but also process layer and information. So, E-shop and Mobile

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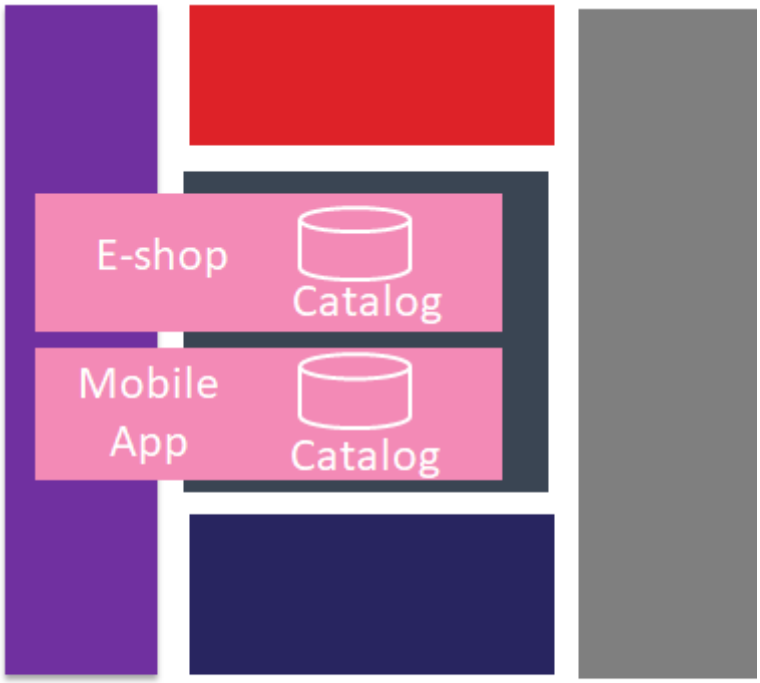
App often have their own catalogue, and global consistency between these catalogues is not always easy to manage.



Legacy usual approach

With ODA, we propose to manage in Core Commerce a global Product Offerings and Product Catalogue component, able to serve all the channels, including their specificities. For example, offers or operations can be allowed only for a subset of channels. E-shop and Mobile App components will query this catalogue and manage its presentation as they want to.

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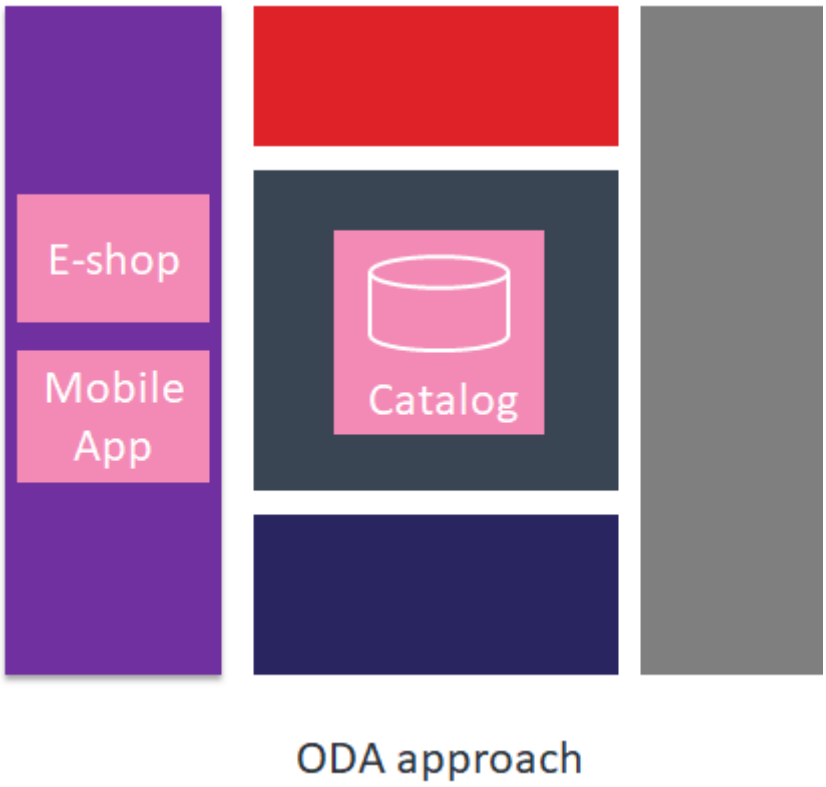
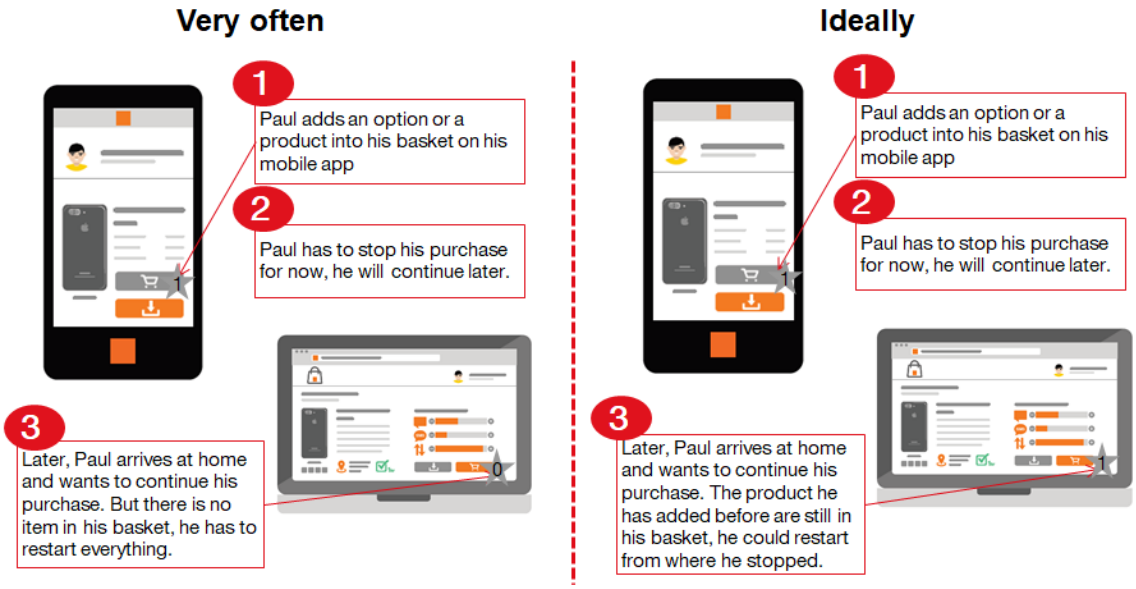


4.1.2. Multi-channel seamless ordering Use Case

A person starts to order offers and products on a channel, using for example the Mobile App provided by a CSP, and needs to stop before the validation of his basket. Back home he wants to continue his order using the CSP Website. Often he is not able to recover his basket and needs to restart his order.

CSPs want to be easily able to provide their customers with a multi-channel seamless ordering.



The reason why this multi-channel seamless ordering is difficult to provide is that CSPs often have implemented applications or components dedicated to one channel and covering GUI but

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also process layer and information. So, E-shop and Mobile App often have not only their own Catalogue but also their own implementation of the Order Capture process. This prevents the sharing of Customer Order information and Order Capture progress and status between the channels.

Legacy usual approach

With ODA, we propose to manage in Core Commerce a global Order Capture process and component, able to serve all the channels, including their specificities, and driven by the Product Offering and Product Catalogue. So, E-shop and Mobile App components will address this Order Capture process, answer its questions, and manage presentation of the basket content. With this decoupling the customer will be able to change channel without losing information, and moreover CSPs will be able to manage easily the support of a CSR if needed : the CSR will directly share the basket initiated by the customer.

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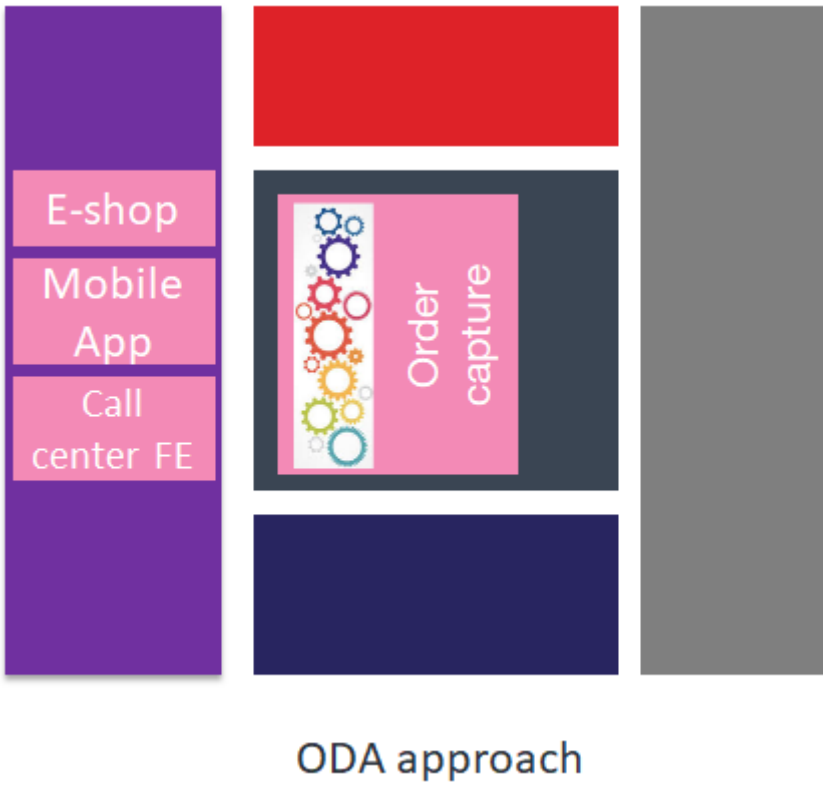
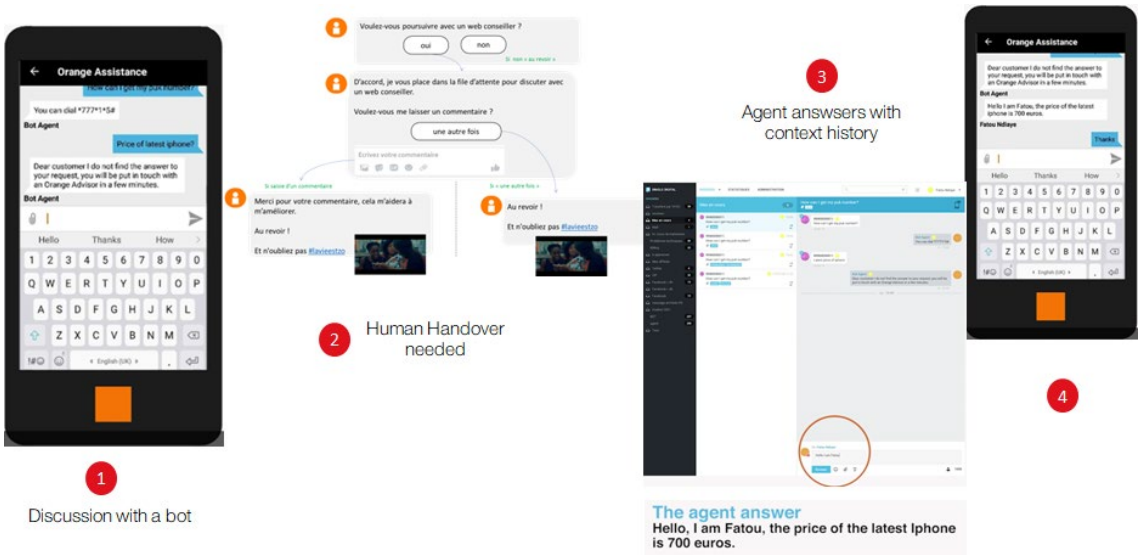


4.1.3. Multi-channel Customer Interaction management Use Case

A person starts to interact with a CSP bot on a channel, for example using the Mobile App provided by the CSP. The bot is able to answer his first questions but at one point a human handover is needed, and a CSR is identified to continue the discussion. This CSR needs to have the context and history of the discussion with the bot to give efficient answers.

CSPs want to be easily able to provide their customers with a multi-channel seamless interaction management.



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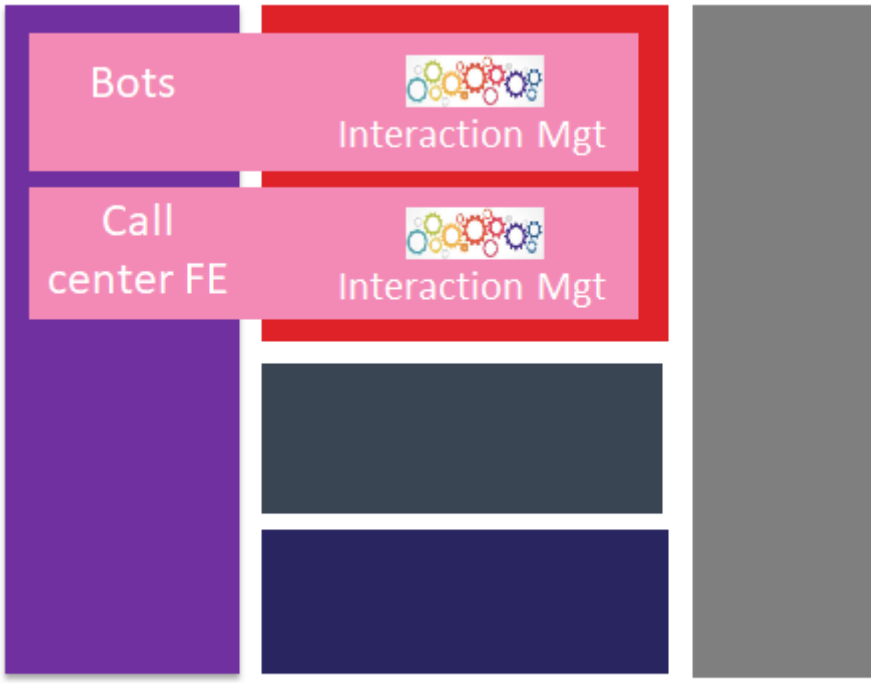
The reason why this multi-channel seamless interaction management is difficult to provide is that CSPs often have implemented applications or components dedicated to one channel and covering GUI but also process layer and information. So, Bots and Call Center often have their own implementation of the Interaction Management process and information. This prevents the sharing of Customer Interaction information and Interaction Management progress and status between the channels.



Legacy usual approach

With ODA, we propose to manage in Party Management a global Interaction Management process and component, able to serve all the channels and to interacts with Bots Knowledge Center in Intelligence Management. So, Bots and Call Center components will address this Interaction Management process, transmit the customer questions, and manage presentation of the answers. With this decoupling the customer will be able to change channel without losing information, and the CSR will directly share the previous questions and answers managed by the bot.

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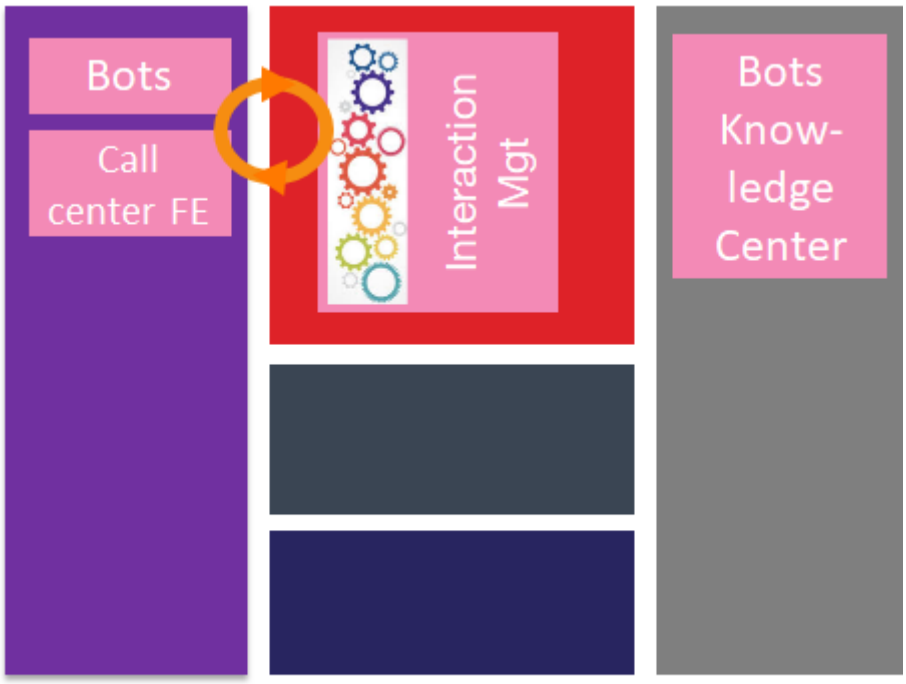

ODA approach

4.2. Multi-Services Identity Management User Story

4.2.1. Seamless Customer Journey Use Case

CSPs are commercializing new lines of business and introducing new products and services, such as Media and Entertainment (e.g., TV, On-demand Streaming Services like Podcasts, Radio etc.), Banking (e.g., Payments, CICO* etc.), Healthcare (e.g., Consultation, Monitoring etc.)

To manage these new lines of business and their services, they require ways to engage with customers and partners which usually leads to implementing dedicated applications with their front-ends. This leads to disconnected experiences that often impact on customer engagement (acquisition, ordering etc.), service abandonment and back office operating efficiencies among others.

Customers and Partners prefer to have single interfaces, that leverage single sign-on for frictionless application use. They also require the ability to from a single engagement interaction, the means to manage their contracts, offers and services without having to switch applications altogether. This can improve customer acquisition, reduce cart abandonment, and reduce customer service and back office support overheads.

* CICO: Cash-in Cash-out

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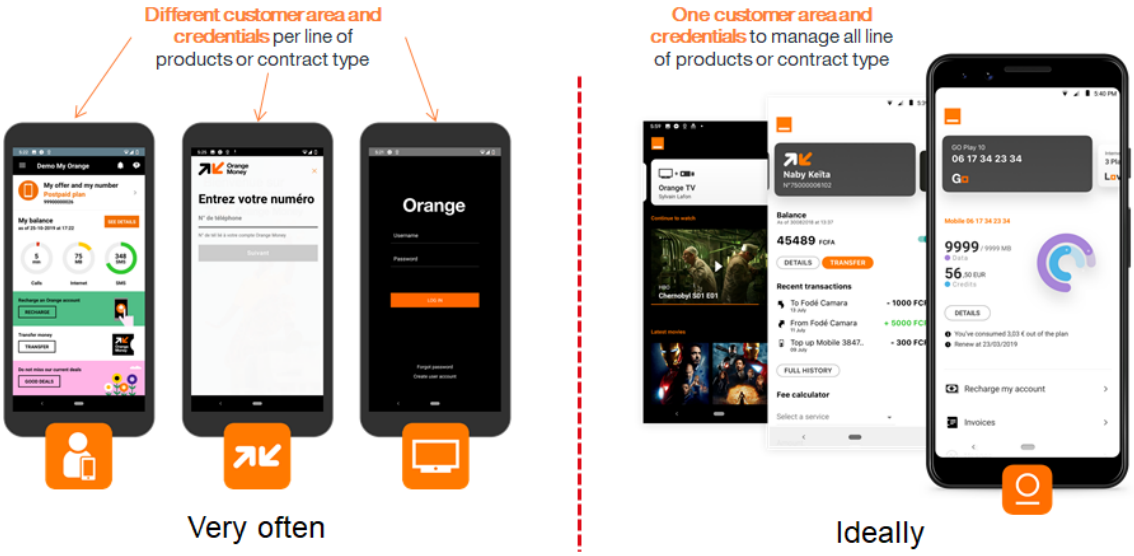

The reason why CSPs often propose dedicated applications and identification / authentication means is that their IT is organized by lines of business, even at the customer level. So, for each contract subscription they register a new customer and give him a new identification / authentication mean to manage this contract.

In this context, it can be even difficult for the CSP to identify all the contracts the same person has subscribed. It can also be difficult to respect some commercial rules, such as Bank regulation. For example, the same person may not be allowed to have 2 Orange Money services, but if he gives 2 different phone numbers to identify himself during the subscriptions the CSP may not be able to stop it the second time. Another example is that in some countries the same person is not allowed to have more than 3 SIM cards. So, if the CSP has an IT for prepaid mobile contracts and another IT for postpaid mobile contracts it is not easily possible to check this rule.

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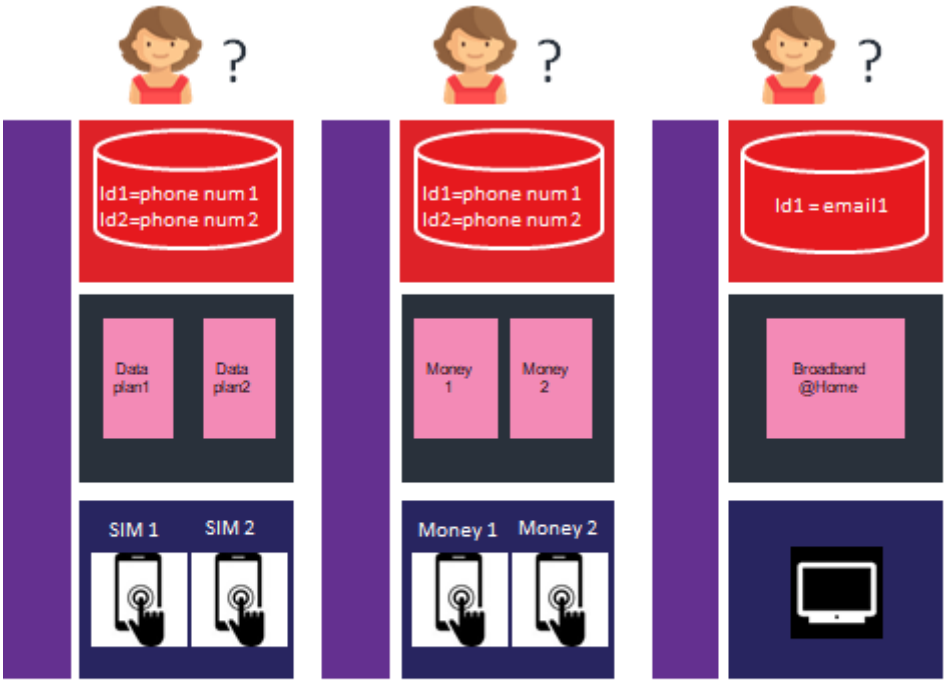

Legacy usual approach: no SSO and possible fraud according to Bank regulation

With ODA, we propose to manage a global Party Management process and component, able to serve all the lines of business, as long as they stay separated in the CSP IT. Associated to Party Roles and Rights management this component will permit to identify a person once, to recognize him during each new contract subscription, and to manage the whole set of the person's roles and rights. An identification / authentication mean will be associated to the person, so the person can exercise all granted rights. Other identification / authentication means this can still be associated to the same person if we want to manage a higher level of security for specific rights for example.

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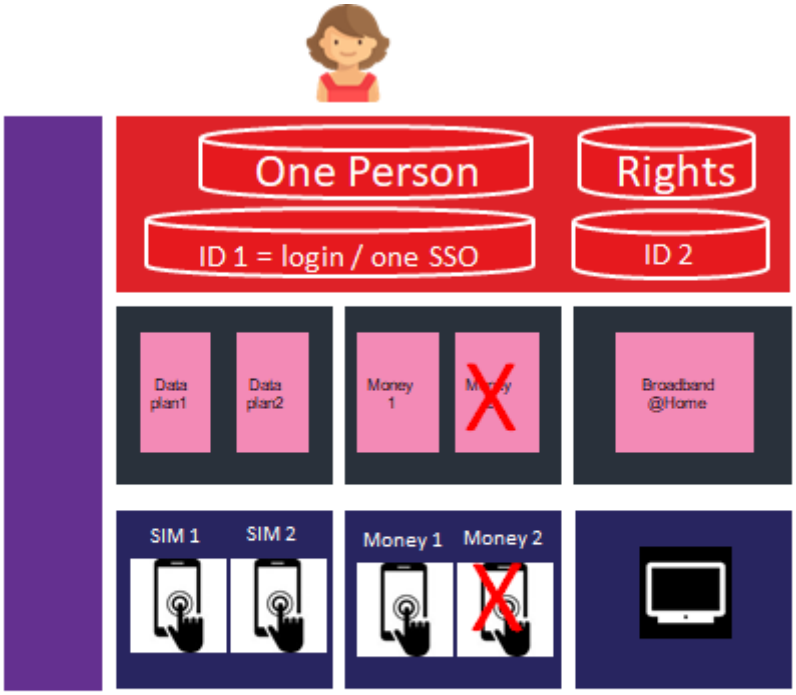
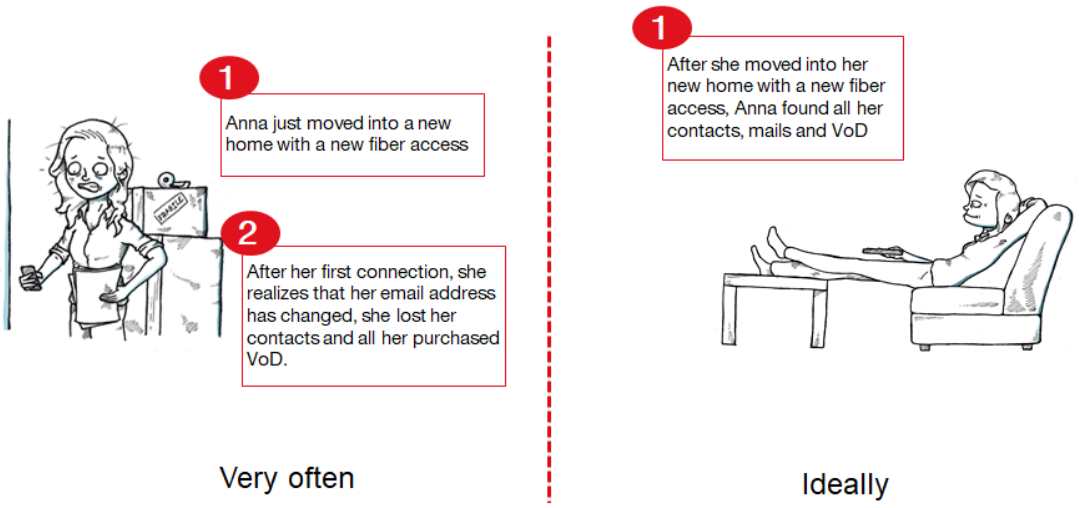


ODA approach: fluid interaction (SSO) and I am recognized as one person

4.3. Seamless Customer Move User Story

4.3.1. Fixed Access Use Case

One of the worst experiences a customer can have with his CSP is to lose his contents (contacts, mails, VOD, ...) when he moves and has to order the installation of a new fix access.



The reason why it can happen is that CSPs not always clearly separate access and services in their Order and Delivery IT. So, when a customer moves, his previous access is deleted, and as many services are managed has linked to this access, they can also be deleted. And when the

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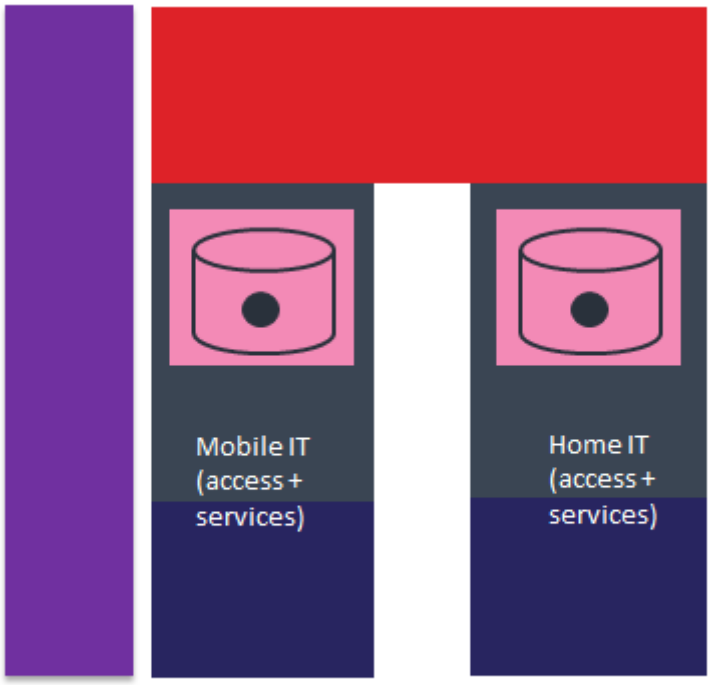
new access is created, new services are too, such as mail or TV/VOD service, loosing existing contents such has contacts or previously purchased VOD.



Legacy usual approach

With ODA, we propose to manage in Core Commerce a global and convergent Offer and Product catalogue, able to describe any type of product and operations. So, we can create a new contract with a new access product corresponding to the new address of the customer and move existing products (or even offers if allowed) to this new contract without any impact at delivery level (pure commercial operation). At production level we also propose to clearly separate systems in charge of access delivery and usage and systems in charge of OTT services.

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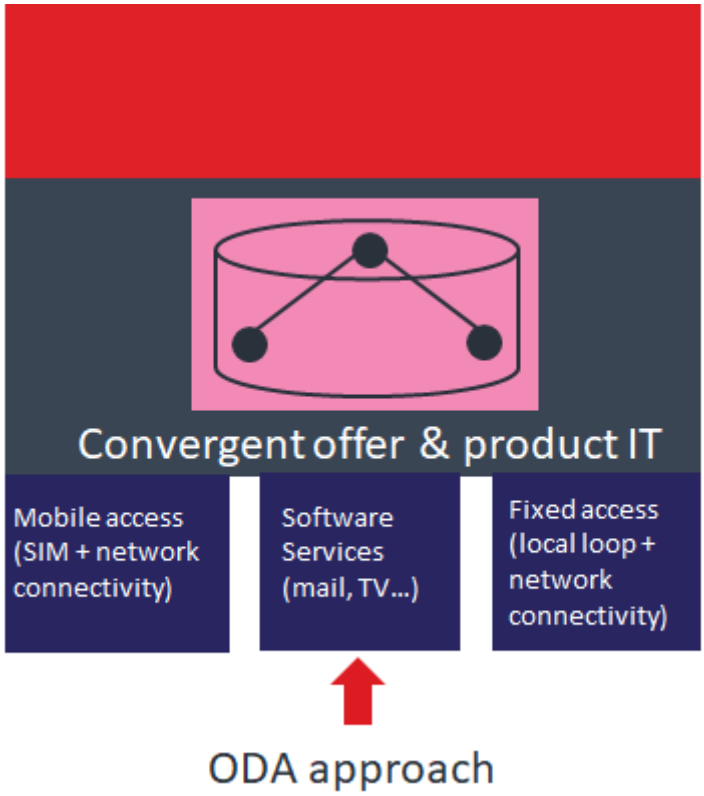

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5. OPEN APIS REQUIREMENTS

In ODA, operational processes are described in the three horizontal blocks (i.e., Party Management, Core Commerce Management and Production). The vertical Engagement Management is in charge of managing interactions with users through the appropriate channel or device with the proper ergonomics.

To allow interactions and decoupling between Engagement Management, in charge of GUI layer, and the other ODA blocs in charge of process layers, Engagement Management is only able to query data, using GET operation provided by standard Open APIs (or POST operation when used for complex queries). And only Party Management, Core Commerce Management and Production are allowed to create/update/delete business information they are in charge, using POST and PATCH operations, because these operations require knowledge of the processes and business rules.

So, 2 new APIs have been defined to permit decoupling, and they can be used to manage articulation between process layers and GUI kinematics, especially for manual tasks:

- TMF701 - Process Flow API,
- TMF688 - Event Management API

5.1. GUI and Business Process Decoupling

The figure below illustrates the location of the current TM Forum Open APIs and the new APIs that permit to decouple GUI and process layers.

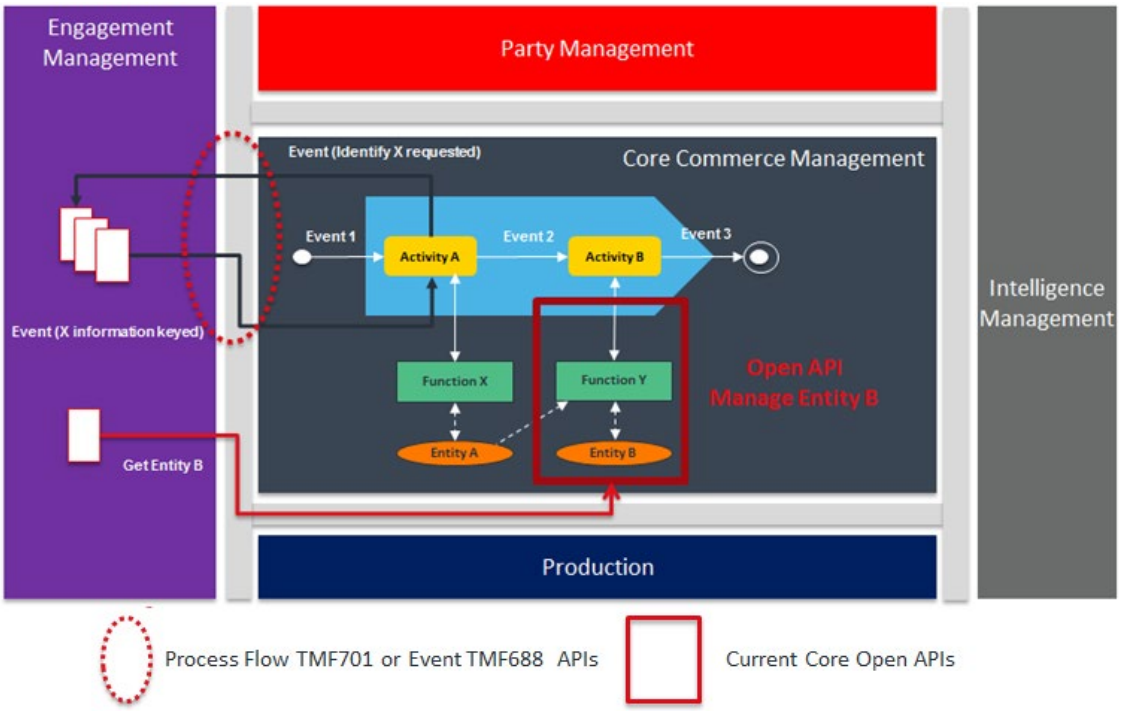

Figure 5.1. ***API Requirements - GUI and Process layers decoupling***

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5.2. Process Delegation and Decoupling

These 2 new APIs can also be used to allow decoupling between processes managed by different ODA blocs and manage an end-to-end business process without duplication / replication of processes, business rules or information.

For example, during an end-to-end Order Capture and Validation process we often need to create or update Party or Billing Account information, to manage payment, ... The Order Capture and Validation process will be globally managed by Core Commerce Management and it will delegate to Party Management the sub-processes related to Party, Billing Account, Payment, ... management.

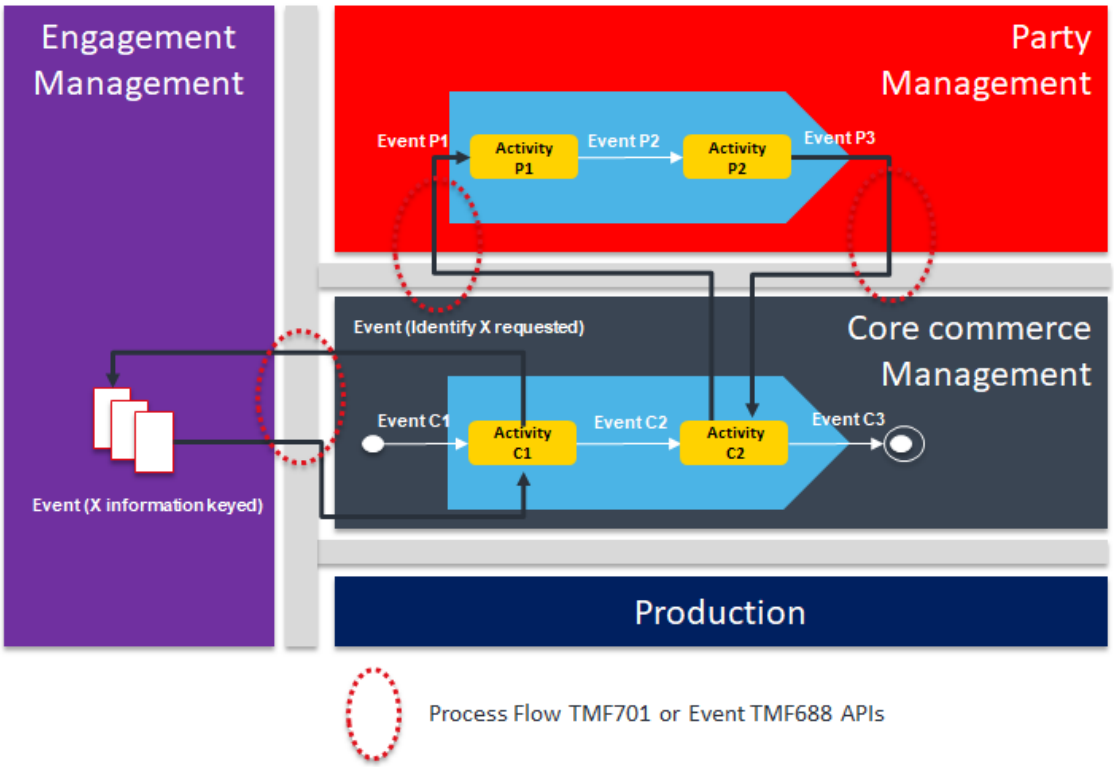

Figure 5.2. ***API Requirements - Process decoupling***

Please refer to IG1228 and its concrete use cases showing through sequence diagrams how TMF Open APIs, including TMF701 and TMF688 can be used.

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6. APPROACH TO MAP BUSINESS INFORMATION OBJECTS TO ODA

The section provides a step-by-step approach to identify Functions covered by an ODA functional block and related business information assets eTOM, SID ABEs.

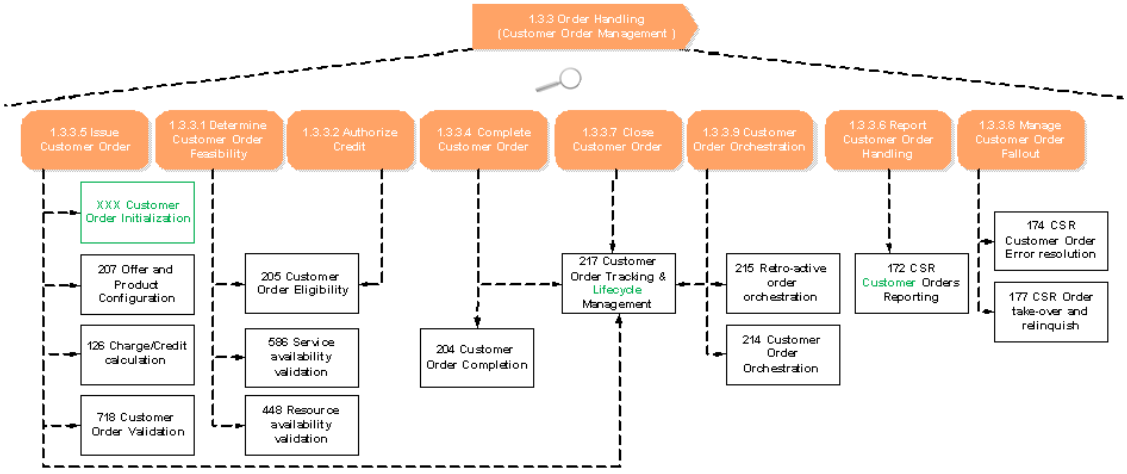
The approach uses:

- well-defined and relevant eTOM and SID mappings described for each ODA functional block (refer to GB1022 ODA Functional Architecture Guidebook);
- the Functional Framework as source of Function names and definitions; and
- links between eTOM business activities, SID ABEs and Functions of the Functional Framework (as these links are not yet fully available, this document helps to initiate or validate them)

6.1. Illustration - Using 'Order Handling' Business Process

6.1.1. Step 1 - Identify all the functions needed by the process

To identify the functions covered by an ODA functional block, we start, for each eTOM level 2 covered, by using the eTOM / Functional Framework mapping (produced by the Frameworks team - ongoing work) and that represents for each eTOM Level 3 business activity which are the functions of the Functional Framework that support this business activity.



Legend: function names in green represent proposed updates of current content of the Functional Framework.

Figure 6.1. ***eTOM / Functional Framework mapping example - Order Handling***

In this example we see that:

- a business activity can be supported by one or several functions
- a function can support one or several business activities

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Then we can identify related business activities, that may be covered by other functional blocks, needed by the end-to-end process. For example, during an order capture process we often need to trigger a payment process. So, the functions that support this payment process will also be identified in the scope.

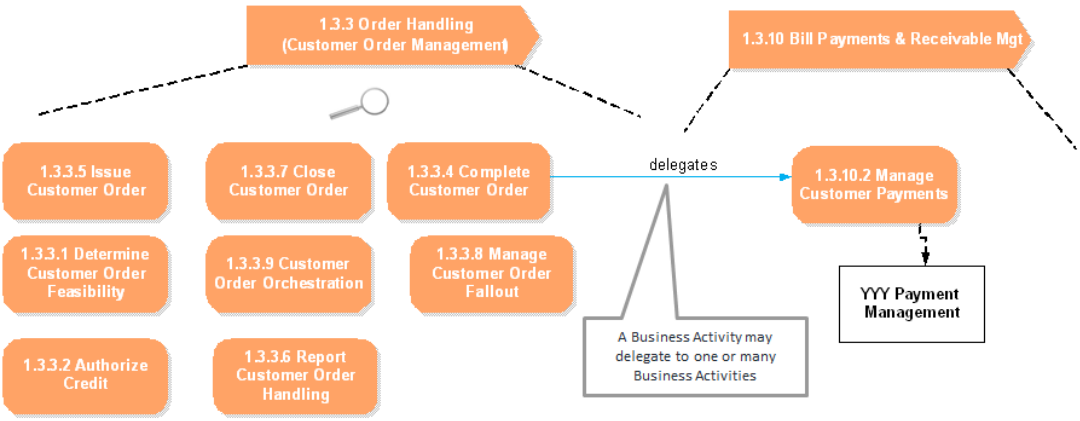

Figure 6.2. ***eTOM business activities delegation example - Order Handling and Payment***

6.1.2. Step 2: Analyze information used & produced by the functions

As a second step, for each function identified, we use the Functional Framework / SID mapping (produced by the Frameworks team - ongoing work) to show which are the SID aggregate business entities (ABE) produced or consumed. And we use the ODA / SID mapping to show which ODA functional block is responsible for each of these ABEs.

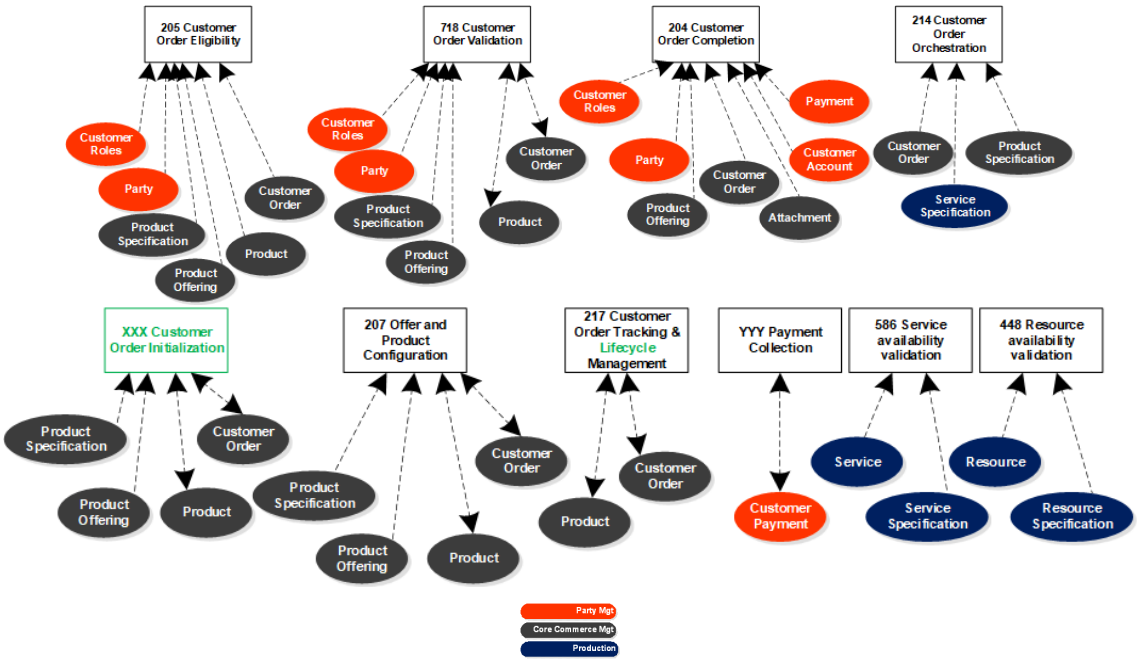

Figure 6.3. ***Functional Framework / SID / ODA mapping - Order Handling and Payment***

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Most of the time, a function consumes and produces information (SID ABEs) positioned in the same functional block as the function itself. But sometimes, a function might consume information in another functional block.

6.1.3. Map the functions to ODA functional blocks

Each function is mapped to the same ODA functional block as the information it produces.

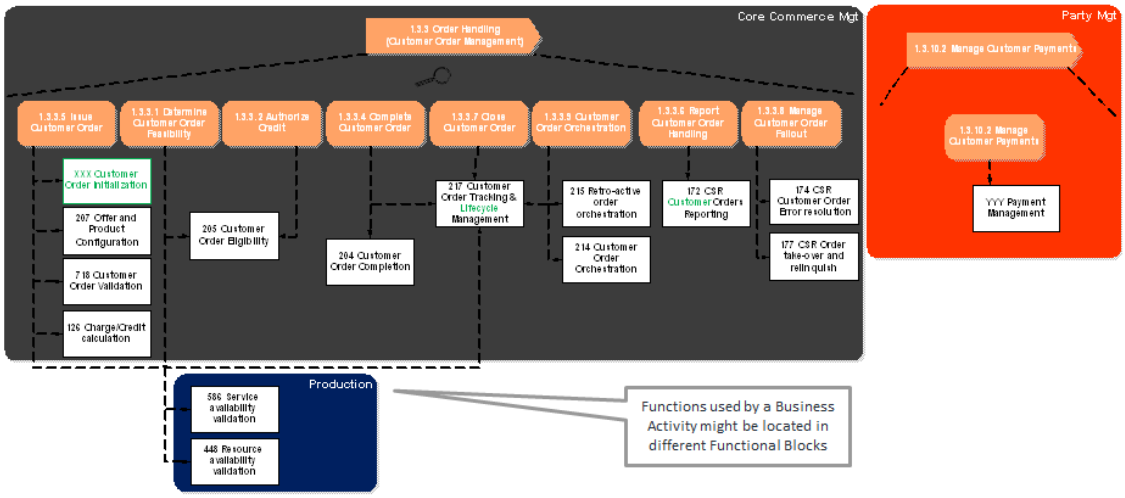

Figure 6.4. ***Functional Framework / ODA mapping - Order Handling and Payment***

From a Business Process such as Order Handling positioned in Core Commerce Management, most of the functions that support it are also managed by Core Commerce. But it can also need Functions (e.g.,: Service Availability Validation), or even complementary processes (e.g., Manage Customer Payments) that are managed by other Functional blocks.

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7. NEW BUSINESS INFORMATION OBJECTS

The section captures the proposals to extend or advance the Functions. Each section captures the new proposals.

7.1. Decoupling & Integration

7.1.1. New Functionalities

This section introduces new changes and proposals that are relevant to extend or advance the 'Decoupling & Integration' function. For the currently adopted standards for 'Decoupling & Integration' functionalities, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0.

Table 1. ***Decoupling & Integration key functionalities***

Functionality	Description / Purpose	Change Reason
TBD	...	
...		

7.2. Engagement Management

7.2.1. New Engagement Management Block functionalities

This section introduces new proposals and changes that are relevant to extend or advance the 'Engagement Management' function. For the currently adopted standards for 'Engagement Management' functionalities, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0

A list of Engagement Management functionalities identified as part of the ongoing study are listed below:

Table 2. ***New Functionalities provided by the Engagement Management Block under review for confirmation include:***

Functionality*	Description / Purpose	Change Reason
Authentication and Authorization	Correctly authenticates and authorizes a Party, including role management.	Separate these functionalities and allocate them to the right function boxes. Current proposals are to split across Engagement Management and Part Management.

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Functionality*	Description / Purpose	Change Reason
		These functionalities correctly authenticate and authorize a Party, including the kinematics and protocols to verify the identity of a user or process (authentication) and specify access rights/privileges to resources.
User-interaction lifecycle management	Manage Party interactions with the different services (provide consent, configure services, etc.) as well as their relationship with other users (create groups and share services)	
TBD	TBD	
...	...	

**These functionalities are still under review for confirmation.*

Remark : Addition of these functionalities to the Functional Framework has to be discussed with the Frameworks team for alignment as part of standardizing them.

7.3. Party Management

7.3.1. Party Management Block functionalities

This section introduces new proposals and changes that are relevant to extend or advance the 'Party Management' function. For the currently adopted standards for 'Party Management' functionalities, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0

7.3.2. New Mapping and CRs to eTOM

Since the Party Management mapping is based on a tactical operating model with no categoric business architecture underpinning it, ongoing analysis and review of the currently mapped businesses processes and new ones are provided here for informative purposes only.

This figure below provides a reference for mapping of new eTOM level business processes, or structural changes that apply to the Party Management function block.

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Figure 8-1. **Figure 7.1.** ***New eTOM Business process mapping proposed*** **for Party Management**

Table 3. ***Process Descriptions (Ref to eTOM for additional details)***

eTOM Process	Category	Process Identifier	Brief Description	Change Reason
Sales Development	(2) eTOM Process Type	1.1.5	Develop the Sales support and response for new and existing products, as well as existing and potential customers.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
Selling	(2) eTOM Process Type	1.1.9	Responsible for managing prospective customers, for qualifying and educating customers, and matching customer expectations Managing prospective parties with whom an enterprise may do business, such as potential existing or new customers and partners, for qualifying	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.

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eTOM Process	Category	Process Identifier	Brief Description	Change Reason
			and educating them, and ensuring their expectations are met.	
Customer Experience Management	(2) eTOM Process Type	1.3.2	Perform customer lifecycle mapping, assess customer experience maturity, as well as analyze customer experience metrics.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
Party Agreement Management	(2) eTOM Process Type	1.6.5	Manage the evaluation of agreements with parties, including customers. Initiate and complete business agreements when one or more other parties are involved.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
Party Identification & Authentication	(2) eTOM Process Type	1.6.16	(TBD)	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
TBD	...			

7.3.3. New Mappings and CRs to SID

This section identifies new mapping or proposed Frameworx SID ABEs to the Party Management function block. For the currently adopted standards for SID mappings, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0.

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Figure 7.2. ***New Mappings for proposed SID ABEs to the Party Management Function block***

Table 4. ***Descriptions of newly proposed changes to SID ABEs (Ref to SID for additional details)***

SID Domain	ABE	ABE Definition	Change Reason
Budget	...		
Document	...		
TBD	...		

7.3.4. New Mapping to Functional Framework

Ongoing Study.

7.3.5. New Reference Interfaces

Ongoing Study

Table 5. ***Proposed Standard Interfaces for Party Management block***

Interface Name	Specification / Reference	Change Reason
TBD	TBD	
...	...	

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7.4. Core Commerce Management

7.4.1. New Core Commerce Management Block functionalities

This section introduces new proposals and changes that are relevant to extend or advance the 'Core Commerce Management' function. For the currently adopted standards for 'Party Management' functionalities, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0.

7.4.2. New Mapping and CRs to eTOM

The Core Commerce Management Block mapping is based on a tactical operating model with no categoric business architecture underpinning it. This mapping is provided for informative purposes only, where different Business Architecture shall have an influence on eventual mapping.

This figure below provides a mapping of newly proposed eTOM level 2 business processes to the Core Commerce Management function block.

Figure 7.3. ***New eTOM Business process mapping proposed for*** **Core Commerce Management**

Table 6. ***New Core Commerce Management eTOM Process Descriptions and Change Reasons.***

eTOM Process	Category	Process Identifier	Brief Description	Change Reason
Sales Development	(2) eTOM Process Type	1.1.5	Develop the Sales support and response for new and existing products, as well as existing and potential customers.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business

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eTOM Process	Category	Process Identifier	Brief Description	Change Reason
				processes to follow the principles of ODA.
Selling	(2) eTOM Process Type	1.1.9	Responsible for managing prospective customers, for qualifying and educating customers, and matching customer expectations Managing prospective parties with whom an enterprise may do business, such as potential existing or new customers and partners, for qualifying and educating them, and ensuring their expectations are met.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
Party Agreement Management	(2) eTOM Process Type	1.6.5	Manage the evaluation of agreements with parties, including customers. Initiate and complete business agreements when one or more other parties are involved.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
Party Special Event Management	(2) eTOM Process Type	1.6.14	Plans, prepare, and produce a special event which targets one or more types of parties. It includes the assessment, definition, acquisition, allocation, direction, control, and analysis of time, finances, people, products, services, and other resources to achieve an event's objectives.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
...				

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7.4.3. New Mapping and CRs to SID

This section identifies new mapping or proposed Frameworx SID ABEs to the Core Commerce function block. For the currently adopted standards for SID mappings to the 'Core Commerce Management', please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0.

Below is the mapping for newly proposed changes to Frameworx SID ABEs to the Core Commerce Management function block.

Figure 7.4. ***New Mapping for proposed changes to SID ABEs to Core Commerce Management***

Table 7. ***New SID ABEs and their description and Change Reasons***

SID Domain	SID ABE	ABE Definition	Change Reason
Opportunity	TBD		
...			

7.4.4. New Reference Interfaces

Ongoing Study

Table 8. ***Proposed new Standard Interfaces for Core Commerce Management block***

Interface Name	Specification / Reference	Change Reason
Proposals requested	TBD	
...	...	

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7.5. Production

7.5.1. New Mapping and CRs to eTOM

This section introduces new proposals and changes that are relevant to extend or advance the 'Production' function. For the currently adopted standards for 'Production' functionalities, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0.

The Production Block's mapping is based on a tactical operating model with no categoric business architecture underpinning it. This mapping is provided for informative purposes only, where different Business Architecture shall have an influence on eventual mapping.

Below is a mapping of new eTOM level 2 business process elements to the Production function block.

Figure 7.5. ***New eTOM Business process mapping proposed for*** **Production.**

Table 9. Production eTOM Process descriptions

eTOM Process	Category	Process Identifier	Brief Description	Change Reason
SM&O Support & Readiness	(2) eTOM Process Type	1.4.4	Manage service infrastructure, ensuring that the appropriate service capacity is available and ready to support the SM&O Fulfillment, Assurance and Billing processes	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of

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eTOM Process	Category	Process Identifier	Brief Description	Change Reason
				ODA.
RM&O Support & Readiness	(2) eTOM Process Type	1.5.4	Manage resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfillment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on the capabilities and costs of the individual FAB processes.	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.

7.5.2. New Mapping and CRs to SID

Below is a mapping of Frameworx SID ABEs to the Production function block.

Figure 7.6. New Production Mapping and CRs to SID ABEs

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Table 10. ***SID ABEs and their description (extract from GB922 19.5)***

SID Domain	SID ABE	ABE Definition	Change Reason
Service	...		
Resource	...		
Enterprise	...		
Common	...		

7.5.3. New Reference Interfaces

Table 11. ***Proposal for new Interfaces to Standard in the Production block***

Interface Name	Specification / Reference	Change Reason
Service related:		
XXXXXX API	Open API TMFXXX	
Resource related:		
XXXXXX API	Open API TMFXXX	
Others:		
XXXXXX API	Open API TMFXXX	

7.6. Intelligence Management

7.6.1. New Mappings and CRs to eTOM

This section introduces new proposals and changes that are relevant to extend or advance the 'Intelligence Management' function. For the currently adopted standards for 'Intelligence Management' functionalities, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0.

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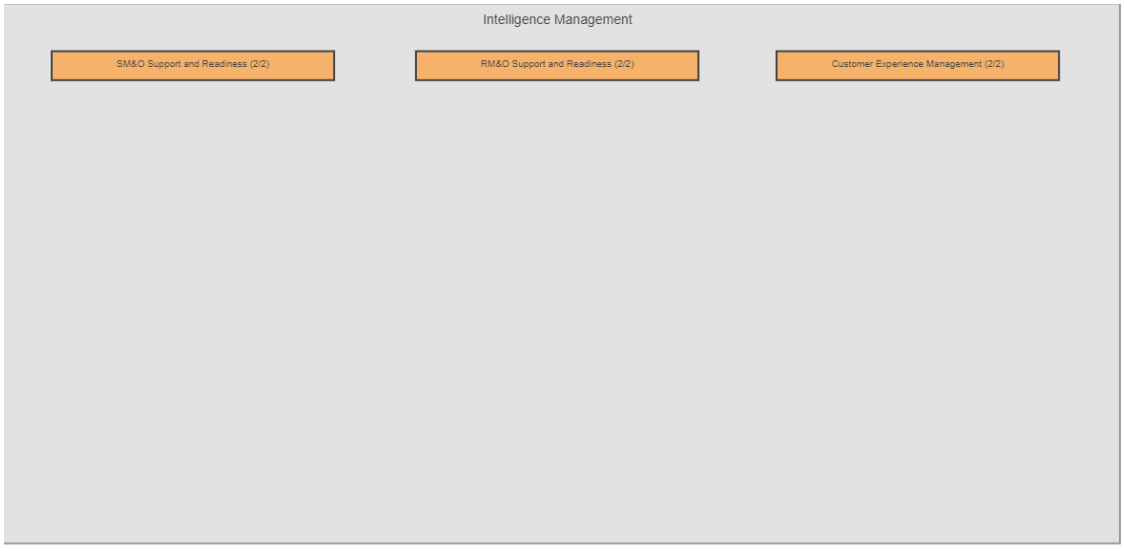

Figure 7.7. ***New eTOM Business process mapping proposed for*** **Intelligence Management**

Note: eTOM level 2 are identified as "split into 2" when they manage SID ABEs that are mapped to different ODA functional blocks.

Table 12. ***Intelligence Management eTOM Process descriptions***

eTOM Process	Category	Process Identifier	Brief Description	Change Reason
SM&O Support and Readiness	(2) eTOM Process Type	1.4.4	Manage service infrastructure, ensuring that the appropriate service capacity is available and ready to support the SM&O Fulfillment, Assurance and Billing processes	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes to follow the principles of ODA.
RM & O Support and Readiness	(2) eTOM Process Type	1.5.4	Manage resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfillment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on	The current Business Process has activities spanning two Functional Architecture blocks. Separation of concerns requires splitting the business processes

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eTOM Process	Category	Process Identifier	Brief Description	Change Reason
			the capabilities and costs of the individual FAB processes. *(Ongoing Study)*	to follow the principles of ODA.
Customer Experience Management	(2) eTOM Process Type	1.6.11		

7.6.2. New Mapping & CRs to SID

Below is a mapping of Frameworx SID ABEs to the Intelligence Management function block.

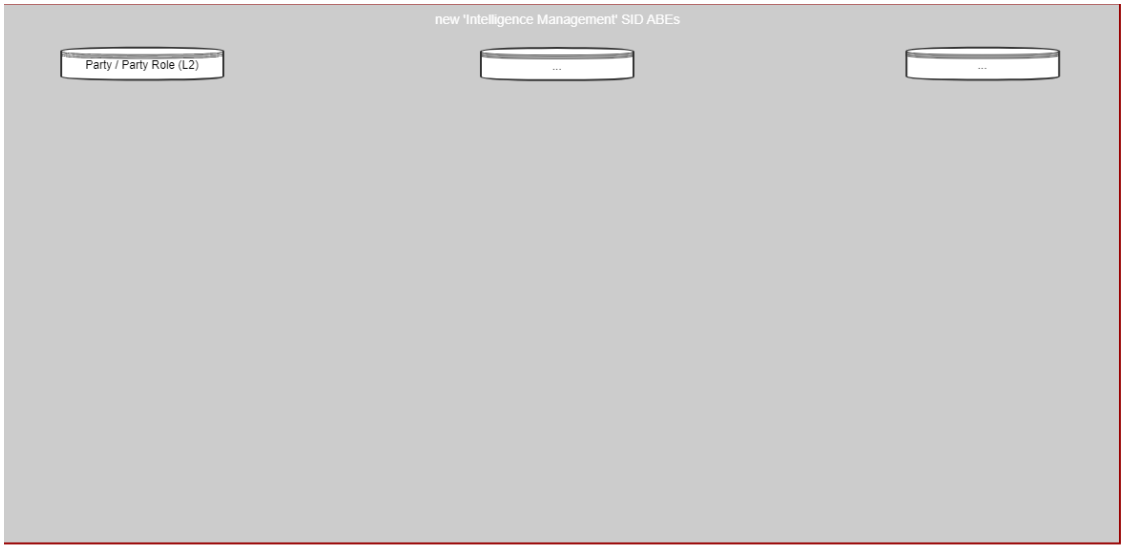

Figure 7.8. ***Mapping SID 19.5 ABE to Intelligence Management***

Table 13. ***Newly proposed SID ABEs and their description***

SID Domain	ABE	ABE Definition	Change Reason
Common	Party / Party Profile (L2)	The characteristics used to group Parties that typify MarketSegments and for the formulation and targeting of MarketingCampaigns. A Party ProfileType is used to categorize one or more PartyRoleSpecifications. A PartyProflieType can be defined for one or more GeographicAreas. The profile can target one or more ProductOfferings and one or more MarketSegments. PartyProfileTypes can be based on (or defined by) DemographicCharacteristics.	

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SID Domain	ABE	ABE Definition	Change Reason
TBD	Market and Sales Performance	*Ongoing Study*	
TBD	Party Performance	*Ongoing Study*	
TBD	Customer Experience	*Ongoing Study*	
TBD	Budget	*Ongoing Study*	
...			

7.6.3. New Functionalities

This section captured a list of Intelligence Management functionalities through various ongoing implementations. For the currently adopted standards for 'Intelligence Management' functionalities, please refer to GB1022 ODA Functional Architecture Guidebook v2.0.0.

Table 14. ***Functionalities covered by the Intelligence Management block***

Functionality	Description / Purpose	Change Reason
RFC		
...		

7.6.4. Capabilities of the Intelligence Management block

The Intelligence Management (IM) Function Block provides “closed-loop” capabilities to package and consume data. It relies on analytical data (reference Analytics Big Data Repository) to develop Insights for decision support, or intent-based actions or advice.

The scope of the Intelligence Management Function Block is considered to be equivalent to that described in ETSI Generic Autonomic Network Architecture (GANA) model specified in ETSI TS 103 195-2. ODA uses GANA for the ODA blueprint model that defines and prescribes the design and operational principles of Decision-Making-Elements (DMEs) called Decision Elements (DEs) in short, as Autonomic Manager Elements.

Modeling for Intelligence Management
Intelligence Management is about Autonomic & Control paradigm that involves Management & Control of Networks and Services.

Most of the time people mix up "Autonomic Management" and "Automated Management". Therefore, within ODA in which a Function Block called "Intelligence Management" Function Block is a key function, a clear definition and scope needs to be settled from the beginning to avoid potential confusion. ETSI GANA (Generic Autonomic Network Architecture) model compared and contrasted both paradigms (Automated Management vs Autonomic Management) and provided a diagram showing responsibility demarcation of each of them

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and how they complement each other as described in ETSI White Paper N0.16 (http://www.etsi.org/images/files/ETSIWhitePapers/etsi_wp16_gana_Ed1_20161011.pdf).

Scope of Automated Management and Autonomic Management & Control

In the table below are the definitions and scopes of both paradigms according to the ETSI GANA model.

Table 15. Definition and scope of diagrams

Automated Management
Automated Management is about workflow reduction and automation i.e., automation of the processes involved in the creation of network configuration input using specialized task automation tools (e.g., scripts, network planning tools, policy generators for conflict-free policies).
Autonomic Management & Control
Citing ETSI White Paper N0.16, is about Decision making Elements (DMEs) or (DEs in short), as Autonomic Functions (meaning Autonomic Manager components that implement the logic that drive the operation of various kinds of control-loops) with cognition capabilities and are introduced in the Management Plane as well as in the Control Plane (whether these Planes are distributed or centralized). "Control" is about control-logic as the kernel of a particular DE that realizes a control-loop in order to dynamically adapt its specifically assigned network resources and parameters or services in response to changes in network goals/policies, context changes and challenges in the network environment that affect service availability and quality.
Autonomic Management & Control vs Automated Management
Automated Management provides input to the Autonomic Management & Control. Indeed, Autonomic Management must exhibit a network governance interface through which the input that governs the configuration of an Autonomic Network should be provided. Thanks to automation tools and mechanisms (Automated Management), by using a high-level language the operator can define the features of the network services that should be provided by the underlying network infrastructure. Such a business language that can help the operator express high level business goals required of the network may be modeled by the use of ontology to add semantics and enable machine reasoning on the goals. The human operator defined features relate to business goals, technical goals and some input configuration data that an Autonomic Network is supposed to use for network resources and parameter configuration.

ODA Intelligence Management organizational model

a. Governance and Management concept outside of the Telecom's sector

Governance related aspect is key issue in all organizations at Government level (e.g., Country Government) as well as at Enterprise level.

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The three most adopted options are:

- Centralized Model
- Distributed Model
- Hybrid Model - combination of both Centralized and Distributed

But the most efficient one is the Hybrid one.

Even the "Autonomic Nervous System" of a human body operates at a "Hybrid structure" that consists of brain in head and the spinal cord with certain "fast reaction" (reflexes) on muscle contractions close to certain lower regions of the human being coordinated in response to stimuli, without necessarily need the need for the central brain's direct involvement during the reflex action.

b. Which option should be adopted by ODA Intelligence Management

Within Digital Service Providers' space there are various processes and life cycles nested and interworking at various time-scales (Fast time scale and Slow time-scale). ODA Intelligence Management has the purpose of managing design and operational processes associated with Autonomic and Cognitive capabilities (Closed Control-loops for management and control of Network and Services) and behaviors. As such, ODA Intelligence Management shall adopt design principles that accommodate and combine both time-scales ("Fast time-scale" and "Slow time-scale") in an interworking manner in a "Hybrid" Model. Hence, this design principle defines time-scaling for such capabilities at various levels of abstraction of self-management functionality.

c. Which available standard can accommodate this ODA Intelligence Management "Hybrid" option

From standardization ecosystem perspective, ETSI GANA (Generic Autonomic Network Architecture) Framework (ETSI TS 103 195-2) is a Hybrid model for Autonomic Management & Control of Network and Services. It is two-level Intelligence Management/Autonomic & Control architecture:

- "macro" Autonomic Managers (responsible for Centralized Closed Control-loops) that drive logically centralized network-wide control-loops but operate at "Slower time-scale".
- "micro" Autonomic Managers that drive similar control-loops that can be implemented in a distributed fashion in Network Elements (NEs) and collaborating with network infrastructure. Within NEs, "micro" Autonomic Managers are for local reactions that operate at "Faster time-scale".

Besides, NGMN and 3GPP SON (Self Organizing Network) architecture standard is based on a Hybrid architecture so called "Hybrid-SON" in the same design architecture principle as the ETSI GANA one. In this regard, we can say the NGMN and 3GPP Hybrid-SON architecture is compliant with the ETSI GANA model as this latter is an agnostic model.

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7.6.5. New Reference Interfaces

The table below lists needed interfaces with the Intelligence Management Function Block (internal interfaces). Each Interface is described and categorized as internal or external.

Table 16. ***New Reference interfaces***

Interface Name	Interface Description	Specification / Reference	Change Reason
RFC			
....			

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8. DEPLOYMENT SCENARIOS

8.1. Engagement Management

(Ongoing Study)

8.2. Party Management

(Ongoing Study)

8.3. Core Commerce Management

(Ongoing Study)

8.4. Production

Production Block Scenarios (with Ecosystem Participants)

While the Production Block is responsible for sourcing and delivering the customer order, outside of handling and shipping physical goods, scheduling workforce for installations, the majority of products will be services exposed, delivered and managed by factories and Operational Domain Management (ODM).

An Operational Domain is typically a scope of management inside an operator (e.g., network operator, service operator) that is delineated by an administrative and/or technological boundary. The technology boundary must be clearly defined as to what services the domain exposes and what resources the domain has exclusive control over.

Types of factories/ODM include:

- OTT Services Management/ Partner Services
- Network services: Physical, virtualized, Hybrid Management
- Cloud SaaS/PaaS/IaaS Management
- Application Management
- Supply Chain Management
- Device Management
- Field Intervention Management

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In addition, an operational domain can also come from external partners (e.g., application provider and network service providers) where clear technological boundaries exist for the external partner. Inside an operational domain, services delivered, and associated infrastructure resources, are managed including exposure and complete lifecycle management of the services delivered from that domain. The lifecycle management includes the life of the service (creation, delivery, removal) and service assurance, usage and related service management.

In some cases, Operational Domains can be classified whether they create services:

- From infrastructure resources, and possibly other operational domain services
- only from other operational domain services (i.e., no direct control over resources or infrastructure)

The former can be considered “technology” operational domains and the latter “composite” operational domains.

An example implementation of these concepts is an Operational Domain Manager (ODM) publishes all information required for the run-time of a network service (NaaS) from the ODM's service catalogue to the CCM product catalog e.g., Optical Transmission service available from 1Meg until 10Gb in increment of 1 Meg. The service provider Product Manager decides to generate 5 products: 1 Meg, 50 Meg, 100 Meg, 1Gb and 10Gb. The flexibility now exists that introducing, say a 7Gb product in the product catalogue would no more involve any changes in the Network factory or ordering system other than passing new attributes. Similarly, introducing a new optical resource from a vendor would not impact the exposed service or OSS as the resource layer is hidden and handled within the ODM.

8.4.1. Factory Model - Orange

This scenario represents the way Orange is setting up their Production area with their factory concepts. Most Factories are autonomous and have their own operational domain(s) that exposes "capabilities" as Customer Facing Services (CFS).

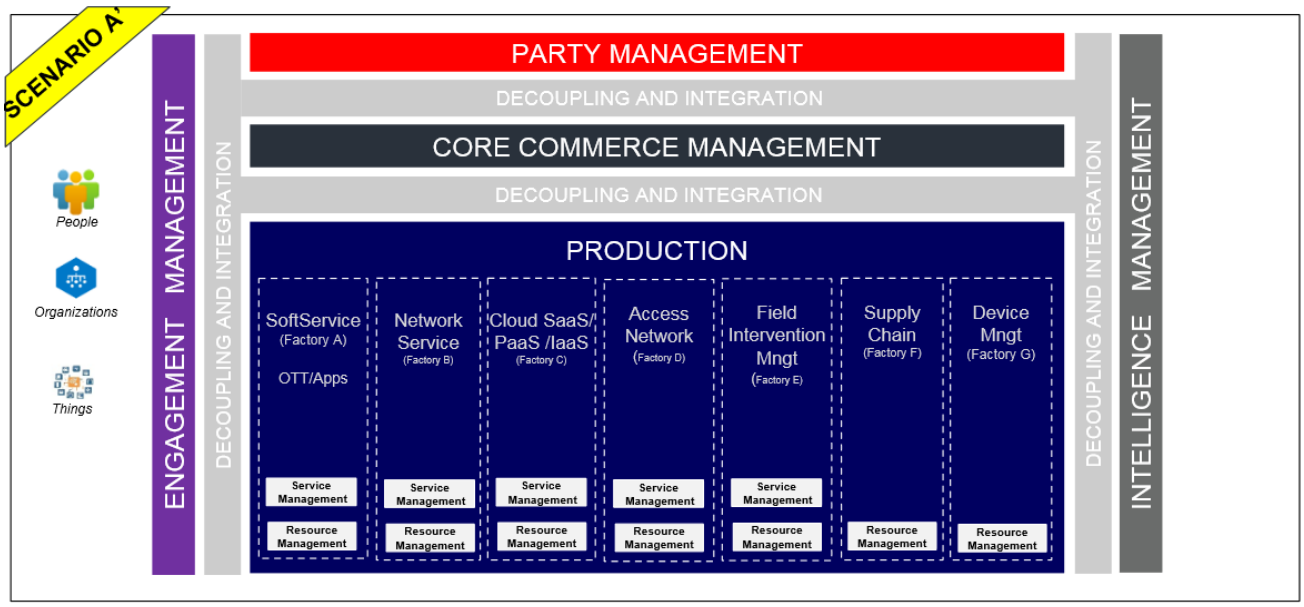

Figure 8.1. ***Orange Production Factory Example***

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In this example, most factories are exposing CFS. Exception is for devices and physical goods, where products cannot be restriction of CFS, so these factories do not expose CFS. Therefore, Service Management is not required for "Supply Chain" and "Device" factories. Also, note the "Access Network" factory is separate from the "SoftService" factory to have the capability to deliver Softservices without any dependency on Access.

Benefits of this proposed deployment approach

The functional architecture is agnostic of any organizational structure, implementation and technical solution, not constrained by existing legacy or technical characteristics. It describes only invariant that can be understood and used by any company telco or not.

Each Factory (SoftService, SaaS ….) provides a type of CFS and have the responsibility of all the processes and data to manage them: Soft Service Factory delivers services like OTT (mail, TV, VoIP), Access factory delivers Access (Fiber, Copper …), Supply Chain delivers goods, FIM delivers human services like installation or training, etc. Moreover, if needed, a factory can request the other factories. e.g., the Soft Service factory needs to instantiate a SaaS, it requests it from the SaaS factory. The access factory may request the FIM to build the access and the FIM can request the supply chain to have the equipment.

With this deployment model:

- There is no need of a transversal orchestration between Factories, because Products are derived from CFS and each CFS is provided by a single Factory. The composition of resources to deliver a CFS is only know by the factory (that may decide to change it)
- There is a real decoupling between Commercial (managed in Commercial Core) and Technical implementation (managed in the Production), with no commercial composition in Production. That is the only way to really reduce TTM.
- Each level (Product, CFS, Technical solution) can have its own lifecycle and evolve at its own speed.
- Each factory can be instantiated or not depending on the needs. e.g., non-incumbent operators do not need an Access Factory, OTT service providers do not know most of the factories and pure retailers even need no factory.
- And it is extendable to any new type of know-how (telco or not) like banking or Health. You just have to add a factory and plug-it to Commerce layer through APIs!

To be understood and used by any company, telco or not, and become a standard, the ODA framework must only describe invariant that are common to everybody. That is what this Functional architecture provides.

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8.4.2. E2E Service Model – Telefonica

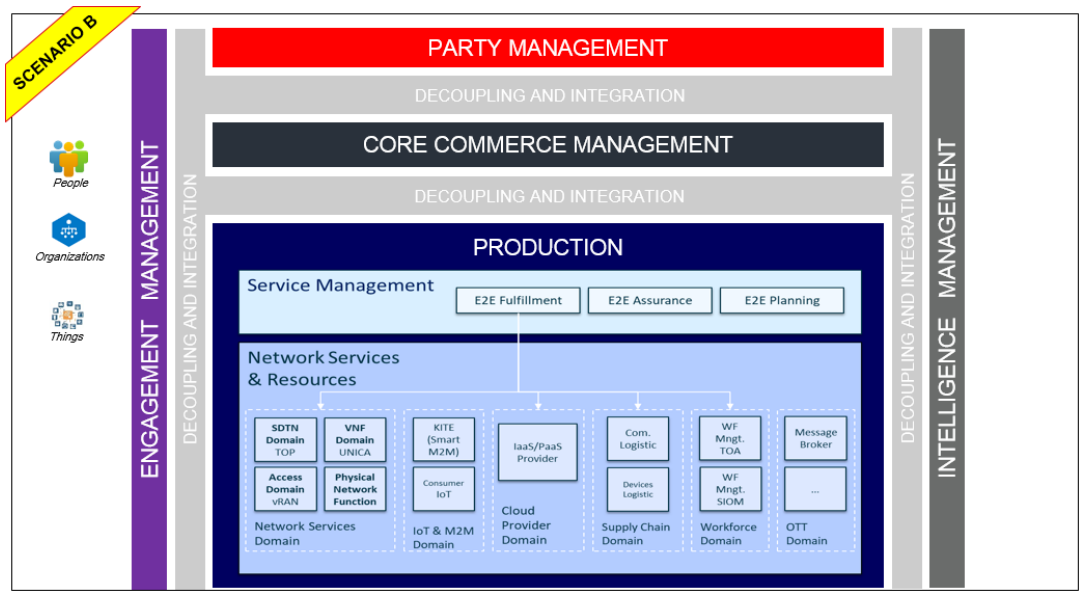

Figure 8.2. ***Telefonica Production Example***

In this example, Telefonica continues to have a "Service Management" layer to handle the technical order through the different domains, proving an end to end control of the different processes (Fulfillment, Assurance and Planning)

Benefits of this proposed deployment approach

- A single E2E view across all resource domains is maintained.
- Reduces Service Management activity duplication
- Implements a single point of Service design irrespective of the Resource domain.
- Service is agnostic of the underlying Resources

8.4.3. NaaS Model – Telstra with Operational Domain Management (ODMs)

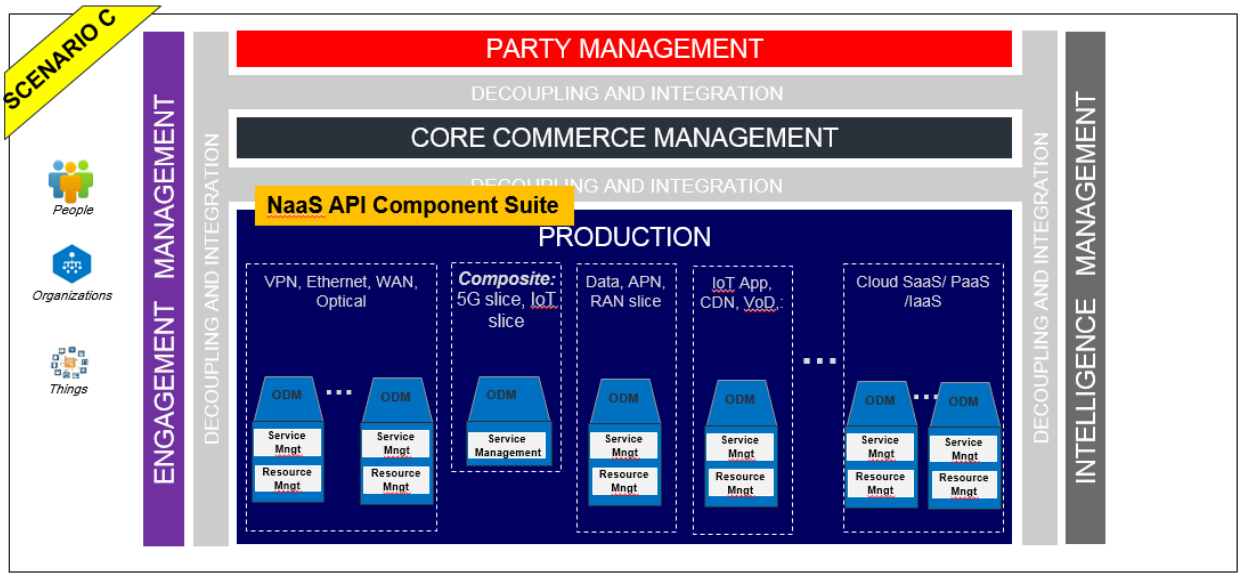

Figure 8.3. ***Telstra Production Example***

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In this example Telstra Networks expose Network as Services (NaaS) grouped by similar functions and/or by organizational level. Every Operational Domain Management (ODM) would be responsible for the entire lifecycle of the services it exposes including the management of resources. Some ODM may be responsible for services that are coordinated across other ODMs and hence would not manage the resources directly. NaaS API component suite (TMF909) is used as the contract of operational functions supported between the Core Commerce Management Block and ODMs as well as between ODM to ODM.

Benefits of this proposed deployment approach

- Expose the network as a service.
- Reduce network management complexity

8.4.4. Microsoft Model - Azure

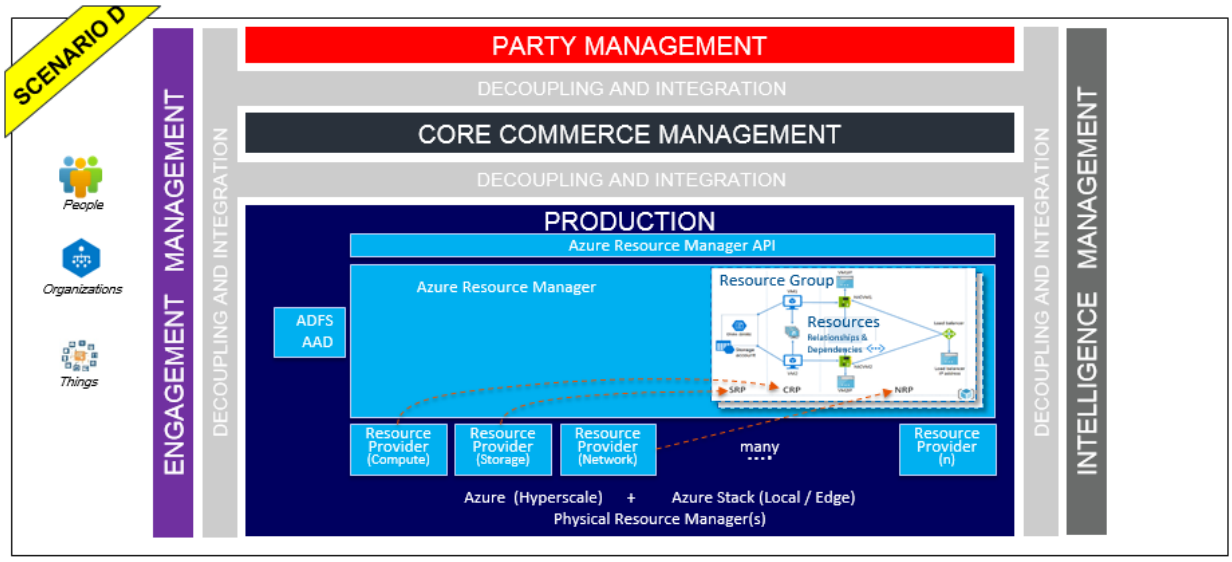

Figure 8.4. ***Microsoft Production Example***

In this example the Microsoft Azure Resource Manager (ARM) is used to model all aspects of application architecture, the relationships, dependencies between service components and all aspects of Operations including Security. The ARM template provides the Azure internal management systems everything needed to deploy and operate any given service or application according to specified policies and specifications. ARM deals with representations of service capabilities and how they are task organized. Hardware resource issues are abstracted and handled separately below the Resource Provider layer by systems not represented here.

Azure is consumed as a Service; a domain of services. The Azure Resource Manager APIs, along with the Azure Marketplace / Store APIs, are organic to Azure and exposed via the Azure Portal, Azure Command Line Interface, and plugins provided to various DevOps tools like Visual Studio or various software orchestrators like Cloudify. Therefore, from a Microsoft point of view, the Azure Resource Manager APIs are as depicted, but they could be said to interact at the Decoupling and Integration layer.

Separately, Azure API Management enables one to publish, map and maintain relationships between native Microsoft APIs and external systems such as ONAP or any particular TM Forum API.

All service levels are set decoratively as developers define application internal architecture, dependencies, and policies via the Resource Group definition. A Resource could be a virtual

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Hard Drive, a Virtual Machine, a VPN, a blob data store, or practically any resource - any SaaS, PaaS, IaaS service. For instance, Azure Kubernetes Service is a Resource. Another example: one could create a Virtual Network Function called PCF modeling its entire architecture including all Resources, their relationships, dependencies and security all within one or more Resource Group(s). A VES agent could be defined as a resource and added to the PCF Resource Group by the Dev to enable automatic reporting to ONAP DCAE upon Operational deployment.

Service Level Metrics / SLAs are collected and exposed via Azure Operations Management Suite (OMS). The VES agent is another option of course.

It is the responsibility of the Resource Manager to maintain adequate capacity for its offerings. A 3rd party can also develop Azure Resource Managers in which case there is more involvement with underlying resource management processes.

Azure Resource Manager definition of terms:

- *Resource - A manageable item that is available through Azure. Some common resources are a virtual machine, storage account, web app, database, and virtual network, but there are many more.*
- *Resource Group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. See Resource groups.*
- *Resource Provider - A service that supplies the resources you can deploy and manage through Resource Manager. Each resource provider offers operations for working with the resources that are deployed. Some common resource providers are Microsoft.Compute, which supplies the virtual machine resource, Microsoft.Storage, which supplies the storage account resource, and Microsoft.Web, which supplies resources related to web apps. See Resource providers.*
- *Resource Manager Template aka ARM Template - A JavaScript Object Notation (JSON) file that defines one or more resources to deploy to a resource group. It also defines the dependencies between the deployed resources. The template can be used to deploy the resources consistently and repeatedly. See Template deployment.*
- *declarative syntax - Syntax that lets you state "Here is what I intend to create" without having to write the sequence of programming commands to create it. The Resource Manager template is an example of declarative syntax. In the file, you define the properties for the infrastructure to deploy to Azure.*

Resource Manager provides several benefits:

- *You can deploy, manage, and monitor all the resources for your solution as a group (Resource Groups), rather than handling these resources individually.*
- *You can repeatedly deploy your solution throughout the development lifecycle and have confidence your resources are deployed in a consistent state.*
- *You can manage your infrastructure through declarative templates rather than scripts.*
- *You can define the dependencies between resources, so they are deployed in the correct order.*
- *You can apply access control to all services in your resource group because Role-Based Access Control (RBAC) is natively integrated into the management platform. Management Groups add flexibility.*

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- *You can apply tags to resources to logically organize all the resources in your subscription.*
- *You can clarify your organization's billing by viewing costs for a group of resources sharing the same tag.*

See also: Understanding Resource Manager deployment and classic deployment.

8.4.5. Platform-based Model - Reference to TR262

Prior ZOOM work on management platform for Hybrid Infrastructure management identified as set of deployment principles which is to realize ODA Functions as platform that exposed group of APIs that address three groups of stakeholders: Customer / Product Manager, Operations Managers, Enterprise Security/Risk Managers.

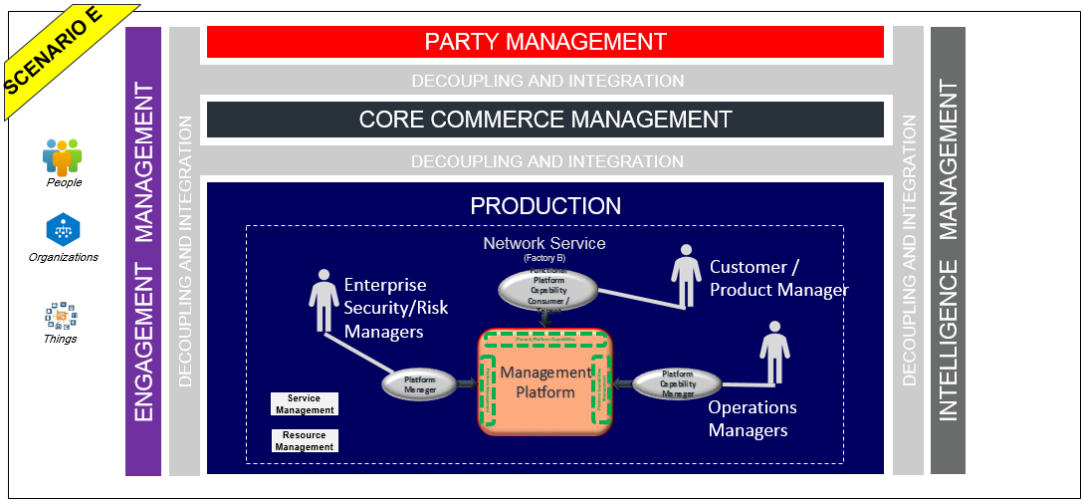

Figure 8.5. ***TR262 Management Platform example***

In this example, a network service is implemented using the Management platform concept described in TR262 Management Platform Blueprint that support the three stakeholder groups: Customer / Product Manager, Operations Managers, Enterprise Security/Risk Managers.

8.4.6. Benefits of this proposed deployment approach

- Support agile evolution of the Management Platform
- Realizes NaaS APIs
- ...

8.4.7. Using ETSI ZSM Model

ETSI ZSM is working on enabling zero-touch automation within the Production Block.

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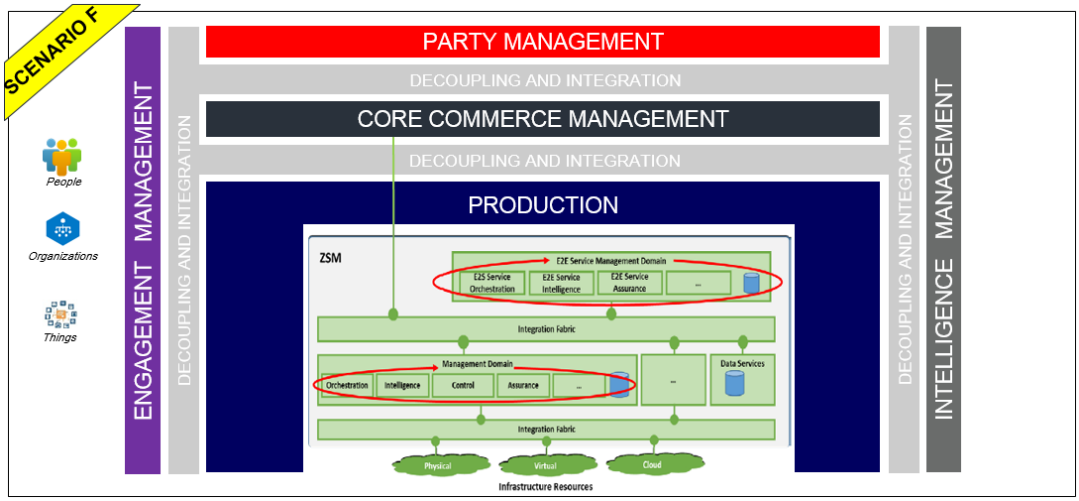

Figure 8.6. ***ETSI ZSM Production example***

In this example the ETSI ZSM “Integration Fabric” is an equivalent to "Decoupling and Integration" Block of the TM Forum ODA Functional Architecture, linking the Core Commerce Management Block with the Production Block. As per this reference, http://www.etsi.org/blog-subscription-information/entry/great-strides-made-by-technical-brainstorming-at-zsm-3 the diagram is work in progress and subject to change. Refer to https://docbox.etsi.org/ISG/ZSM/Open for latest work.

8.5. Intelligence Management

8.5.1. Deployment Scenario 1 - Building AI / Autonomics / Cognitive capabilities are built-in per Design in the Intelligence Management Function

Figure 7-7 depicts the ODA Intelligence Management Framework. In this Framework, AI / Autonomics / Cognitive capabilities are built-in per Design in the Intelligence Management Function Block and in each of the three horizontal ODA Function Blocks. Each having its own “Decision-Making Logic”, its own “local Data Lake” so called Knowledge Base (KB) for the purpose of fast / Distributed / Local decision-making element (d_DE), realized by fast control loops complemented by “centralized” Decision-Making logics for the purpose of a holistic decision-making view (at each ODA horizontal Function Block as a whole) at slow time scale. This can be achieved through slow / centralized control loops and "centralized" Data lake or centralized Knowledge Base that can be located in the Intelligence Management Function Block. The Intelligence Management Function Block is the place where Data Lake / centralized Knowledge Base (cKB) as a "Shared Data repository" used by the three Closed Control loops should be located. It is also the place where “Centralized” decision-making element (Centralized_DE) are located.

Both types of Decision-Making-Elements or Decision Elements (DEs) are either Logically Centralized or Distributed Control Software Logics (DEs) that operate in different time-scales but interworking harmoniously in realizing autonomic behaviors:

- Self-configuration,
- Self-optimization,
- Self- Healing

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- Self-* of Managed Entities

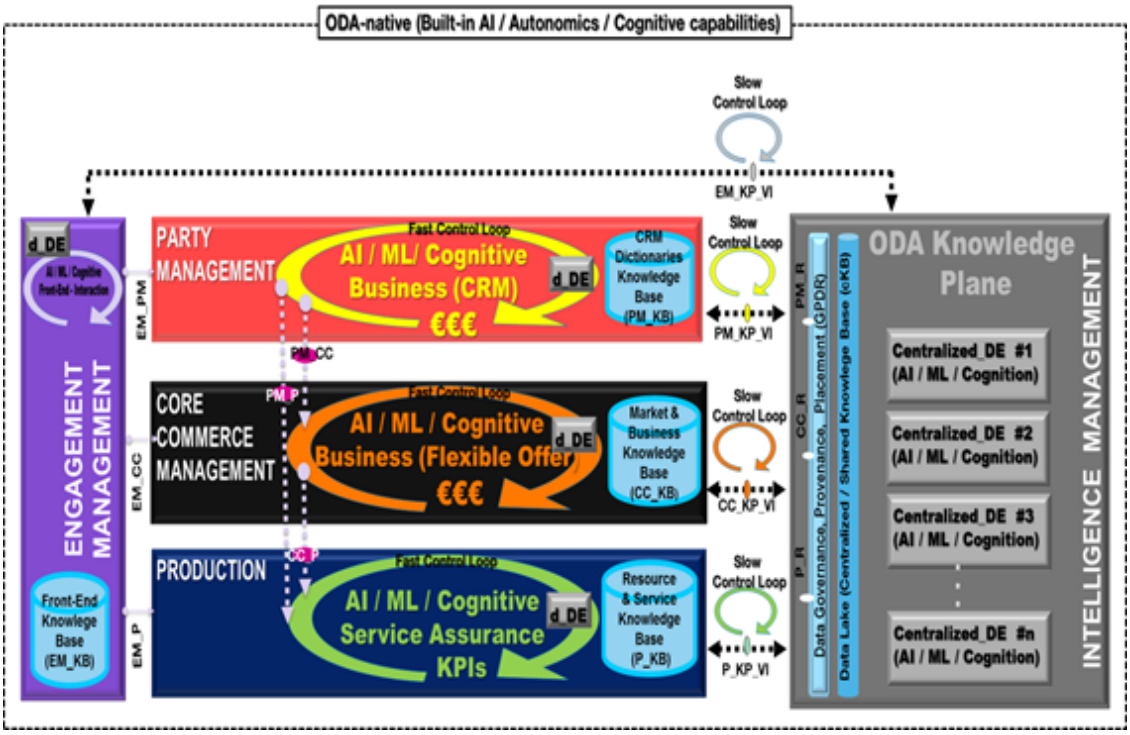

Figure 8.7. ***ODA Intelligence Management Deployment option 1***

Distributed / Local / Fast decision-making element or "d_DE" (per ODA Function Block) design principle

This distributed decision-making logic with less complex Analytics for Autonomics / AI (Cognitive) Algorithms is realized by fast control loops located in each of the three ODA Function Blocks. This is reflected at the left-hand-side of Figure 7-7.

In order to distinguish distributed / local / fast Decision-Making-Elements associated with the three horizontal ODA Function Blocks, we gave them a specific name and color:

- Yellow fast Closed Control loop) or AI / ML / Cognitive Business (CRM): "Party Management Closed Control Loop" with its "Business & Marketing" CRM Knowledge Base and its "Decision-Making Logics" (AI Algorithms e.g., those provided by the industry, …)
- Orange fast Closed Control loop or AI / ML / Cognitive Business (Flexible Offer & Business Assurance) : "Core Commerce Management Closed Control Loop" with its "Business & Marketing" Knowledge Base and its "Decision-Making Logics" (AI Algorithms e.g., those provided by the industry,…)
- Green fast Closed Control loop or AI / ML / Autonomics / Cognitive Service Assurance: "Production Closed Control Loop" with its "Resource & Service" and its "Decision-Making Logics" (AI Algorithms e.g., ETSI GANA Decision-Making-Elements (DMEs) Algorithms)
- Purple fast Closed Control loop or AI / ML / Autonomics / Cognitive (Dynamic Front-End-Interaction): "Engagement Closed Control Loop" with its "Front-End" Knowledge Base and its "Decision-Making Logics" (AI Algorithms e.g., those provided by the industry…)

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Interworking between Closed Control loops and concept of Delegations-Escalation-Synchronization design principle

Those four Closed Control loops (their associated Decision-Making-Elements) may interact to fulfill business and operations objective. Therefore, they can be configured hierarchically (nested) and each control-loop's operation is characterized by a specific life cycle.

- Those four Closed Control loops shall interact in a coordinated manner per design to prevent any conflicts that may happen. Therefore, Delegations-Escalation-Synchronization rules along with mapping tables (relationships between Decision-Making-Elements and associated policy-controlled Managed Entities) shall be specified to ensure a stable interworking of the control-loops for achieving a common efficient decision-making objective (conflict-free). Both Top-Down nesting & interworking and Button-up can be considered.
- There is a need for specifying Reference Points /APIs to reflect on those vertical interactions between the four Closed Control Loops categories as depicted in Figure 4-19. The specification of those Reference Points are implementation-specific (see IG1177)

Logically Centralized / Outer / Slow decision-making design principle

This logically centralized / slow decision-making logic with complex Analytics for Autonomics / AI (Cognitive) Algorithms is realized by slow control loops located in the ODA Management Plane or Control Plane associated within each of the three ODA Function Blocks (Party Management, Core Commerce Management, Production Block) respectively.

In order to distinguish logically centralized / outer / Slow closed Control loops associated with the four ODA Function Blocks coloring has been synchronized with ODA FA.

Key Implementation requirements associated with Decision-Making-Elements (DMEs) (for Centralized DMEs and Distributed DMEs) are defined in IG1177.

Synchronization between local Knowledge Bases and Centralized Knowledge Base (cKB) architecture requirements

ODA shall specify an interface at the Intelligence Management Function Block with the purpose to updating and synchronizing Local Knowledge Bases and Centralized Knowledge Bases. It is noted that the Intelligence Management Function Block should be the place where Data Lake / centralized Knowledge Base (cKB) as a "Shared Data repository" used by the three Closed Control loops) is located and managed and made available to be consumed by any authorized "Entity" and audited and verified by the Regulator. Those three Reference Points / Interfaces as shown in Figure 4-19 are subject to specification and IG1177.

- ODA Knowledge Base (Shared Data Repository) shall manage the storage of the content (Data / Knowledge) only and make it available to any authorized consumer through Publish / Subscribe, Query & Find process.
- The processing of Knowledge / Data shall be handled elsewhere in Analytical Modules or in DMEs or both

ODA Knowledge Base and Regulation architecture requirements

ODA shall specify a reference point then make it available to Regulator to allow this Body verifying and auditing Data Governance, Provenance and Placement including GDPR enforcement at each of the three ODA Function Blocks by interacting with their respective Knowledge Bases as well as with "centralized" Knowledge Base (cKB as shown in Figure 4-19).

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8.5.2. Deployment Scenario 2 - Instantiation of ETSI GANA Framework unto ODA Intelligence Management

The scope of the Intelligence Management Function Block is considered to be equivalent to that described in ETSI Generic Autonomic Network Architecture (GANA) specified in ETSI 103 195-2. Figure 7-8 depicts the mapping of ETSI Generic Autonomic Network Architecture (GANA) with ODA Intelligence Management and Production Function Block. This Mapping can involve the other ODA Function Blocks as illustrated in the deployment scenario below.

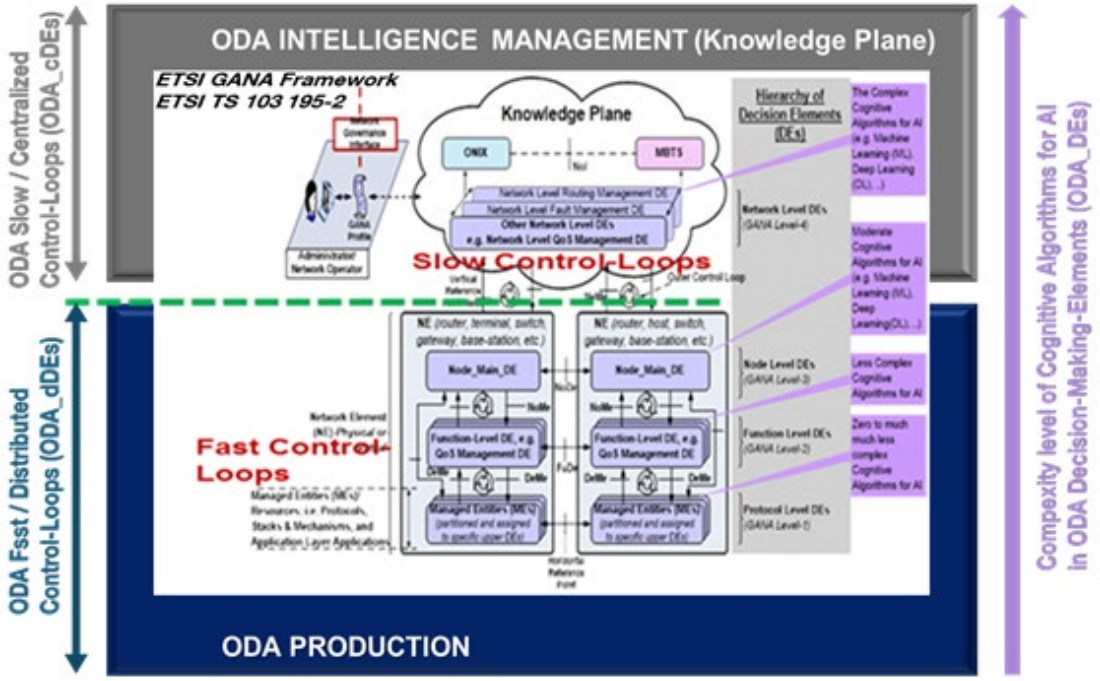

Figure 8.8. ***Intelligence Management Deployment option 2: Instantiation of ETSI GANA Framework onto ODA Intelligence Management***

Table 17. Illustrates Mapping / Instantiation

ETSI GANA Knowledge Plane	ODA Knowledge
• GANA Centralized Closed-Control Loops (cDEs) with Analytics	• ODA Centralized Closed-Control Loops (cDEs) with Analytics
• ONIX (Overlay Network Information exchange Servers)	• ODA Knowledge Base / Shared Data Repository
• MBTS (Model Based Translation Service)	• Translator of data from multiple sources in order to create federated information or data models
GANA Network Element / Managed Entities (MEs)	**ODA Production Element / Managed Entities (MEs)**
• GANA Nodes	• ODA Nodes

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ETSI GANA Knowledge Plane	ODA Knowledge
• Function (Network Function)	• ODA Network Functions
• Protocols / Mechanisms	• ODA Services (Microservices)
• Managed Entity Components	• ODA Components
GANA Network Elements distributed Closed-Control Loops (dDEs)	**ODA Production Elements' distributed Closed-Control Loops (dDEs)**
• Network Element-level d_DE	• Network Element-level d_DE
• Node-level d_DE	• Node-level d_DE
• Function (Network Function)-level d_DE	• Function (Network Function)-level d_DE
• Protocol / Mechanism-level d_DE	• Service (Microservice)-level d_DE

For detailed implementation scenarios, please refer to Appendix A.

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9. ADMINISTRATIVE APPENDIX

9.1. References

9.1.1. Core References

- Open Digital Architecture Whitepaper - https://www.tmforum.org/resources/whitepapers/open-digital-architecture/
- OSS of the Future - https://www.tmforum.org/resources/whitepapers/oss-of-the-future/
- Open Digital Architecture Vision - IG1166 Open Digital Architecture Vision R18.0.0
- Open Digital Architecture Concepts & Principles - GB998 Open Digital Architecture (ODA) Concepts & Principles R18.0.0

9.1.2. Other Related References

- GB1022 ODA Functional Architecture Guidebook v2.0.0.
- IG1176 TOSCA Guide for Model-Driven Automation R19.0.0
- GB999 ODA Production Implementation Guidelines R19.0.0
- IG1130F SDN/NFV impacts on TAM Knowledge Management R19.0
- IG1157 Digital Platform Reference Architecture Concepts and Principles R19.0.1

9.1.3. Terminology

- Open Digital Architecture Terminology - TMF071 ODA Terminology R18.0.0)

9.2. Document History

9.2.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1	31-May-2018	Emmanuel A. Otchere	Initial Version
1.0.0	26-Jun-2018	Alan Pope	Minor edits prior to publication
1.0.1	16-Jul-2018	Adrienne Walcott	Formatting/style edits prior to R18 publishing
1.0.2	30-Nov-2018	Brice Petitfrère	Update of Reference Interface Description, for each ODA block
1.0.3	04-Dec-2018	Sylvie Demarest	Update of eTOM and SID mappings, for each ODA block. Add of SID ABEs definitions, for each ODA block

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Version Number	Date Modified	Modified by:	Description of changes
2.0.0	28-Dec-2018	Alan Pope	Final edits prior to release.
3.0.0	25-Jun-2019	Alan Pope	Final edits prior to release.
3.0.1	17-Oct-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
4.0.0	02-Dec-2019	Alan Pope	Final edits prior to release.
4.0.1	24-Feb-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status
5.0.0	30-Jan-2020	Alan Pope	Final edits prior to publication
5.0.1	24-Mar-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status
5.1.0	28-Jan-2021	Emmanuel A. Otchere	Restructured as an exploratory document for ODA-FA advancement
5.1.0	26-Mar-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status
6.0.0	02-Apr-2021	Alan Pope	Final edits prior to publication

9.2.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
18.0.0	12-July-2018	Alan Pope	Initial Release
18.5.0	28-Dec-2018	Alan Pope	New Release
19.0.0	25-Jun-2019	Alan Pope	New Release
19.0.1	17-Oct-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
Pre-production	02-Dec-2019	Alan Pope	New Release
Production	24-Feb-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status

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Release Status	Date Modified	Modified by:	Description of changes
Pre-production	31-Jan-2020	Alan Pope	New Version
Production	24-Mar-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status
Pre-production	29-Jan-2021	Alan Pope	Updated to version 5.1.0
Production	26-Mar-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status
Pre-production	02-Apr-2021	Alan Pope	Updated to version 6.0.0
Production	24-May-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status

9.3. Acknowledgements

This document was prepared by the members of the TM Forum ODA project:

- Sylvie Demarest, Orange
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- Daniela Galigniana, Telefónica
- Dave Milham, TM Forum
- Eric Troup, Microsoft Corporation
- Johanne Mayer, Telstra
- Tayeb Ben Meriem, Orange
- Cecile Ludwichowski, Orange

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10. Appendix A: Detailed Intelligence Management Implementation Scenarios

10.1. INTRODUCTION

10.1.1. Deployment Flavors / Features / Scenarios / Options | or How to use

ODA Intelligence Management categories
For the four use cases described in this section, we are considering two categories of ODA:

1. Home ODA. It is controlled by a Digital Service Provider (DSP)
2. External ODA. It is controlled by a DSP's Partner

This categorization refers to the concept of ownership defined earlier which sets the responsibility demarcation from an autonomic perspective. It defines how embedded Autonomic / AI / Cognitive capabilities are arranged within the ODA (relationships between DEs and MEs inside a given ODA category).

All Managed Entities considered in this section belong to the Production Function Block of the ODA Functional Architecture. But this principle could apply to the two other horizontal ODA Function Blocks as well.

Use Case 1: Hierarchy of ODAs
This use case illustrates the situation where two ODAs (Home ODA and External ODA) are collaborating from decision-making perspective under hierarchical relationship. This model is built when there is a need for "Delegation - Escalation-Synchronization" operations to prevent conflicts that may happen in such a model.

Figure 10-1 depicts this use case. It is made of two "ODAs":

- External ODA. It contains an "External Autonomic Manager" positioned at Upper / Higher level in the hierarchical model. Its policy-controls the Home Autonomic Manager belonging to the Home Domain.
- Home ODA. It contains a " Home Autonomic Manager" positioned at Lower level in the hierarchical model. It is policy-controlled by the External Autonomic Manager
- This interaction is shown by the yellow arrow.

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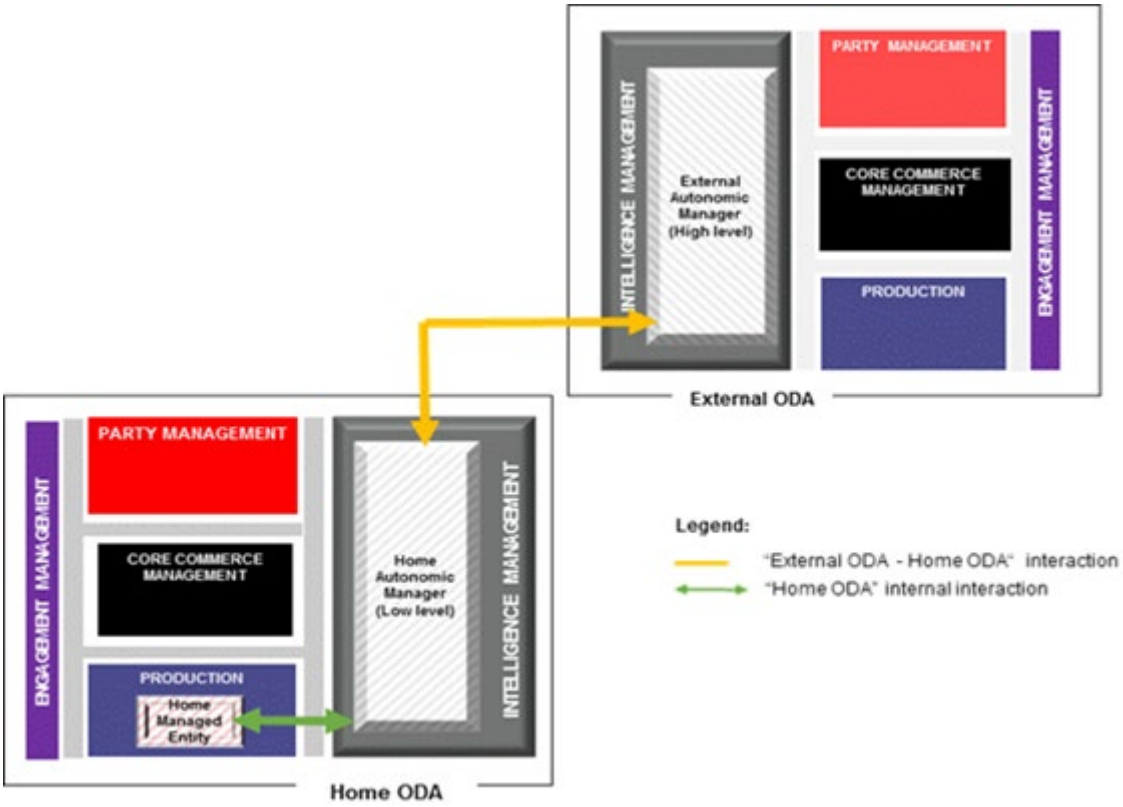

Figure 10.1. Hierarchy of ODAs

Use Case 2: Federation of ODAs (Peering / Sibling model)

This use case illustrates the situation where two ODAs (Home ODA and External ODA) are collaborating from decision-making perspective under federation relationships. This model is built when there is a need for federating ODAs to deliver E2E service, for instance E2E Network Slice crossing two ODAs. Cognitive Fulfillment & Service Assurance of the E2E Network Slice necessitate E2E decision-making cross those two ODAs.

Figure 10-2 depicts this use case. It is made of two "ODAs"

- External ODA. It contains an "External Autonomic Manager" positioned as the same level w.r.t. a Home Autonomic Manager.
- Home ODA. It contains a " Home Autonomic Manager" positioned at the same level as an External Autonomic Manager
- This interaction is shown by the yellow arrow

External Autonomic Manager is seen as a Peer from Home Autonomic Manager perspective.

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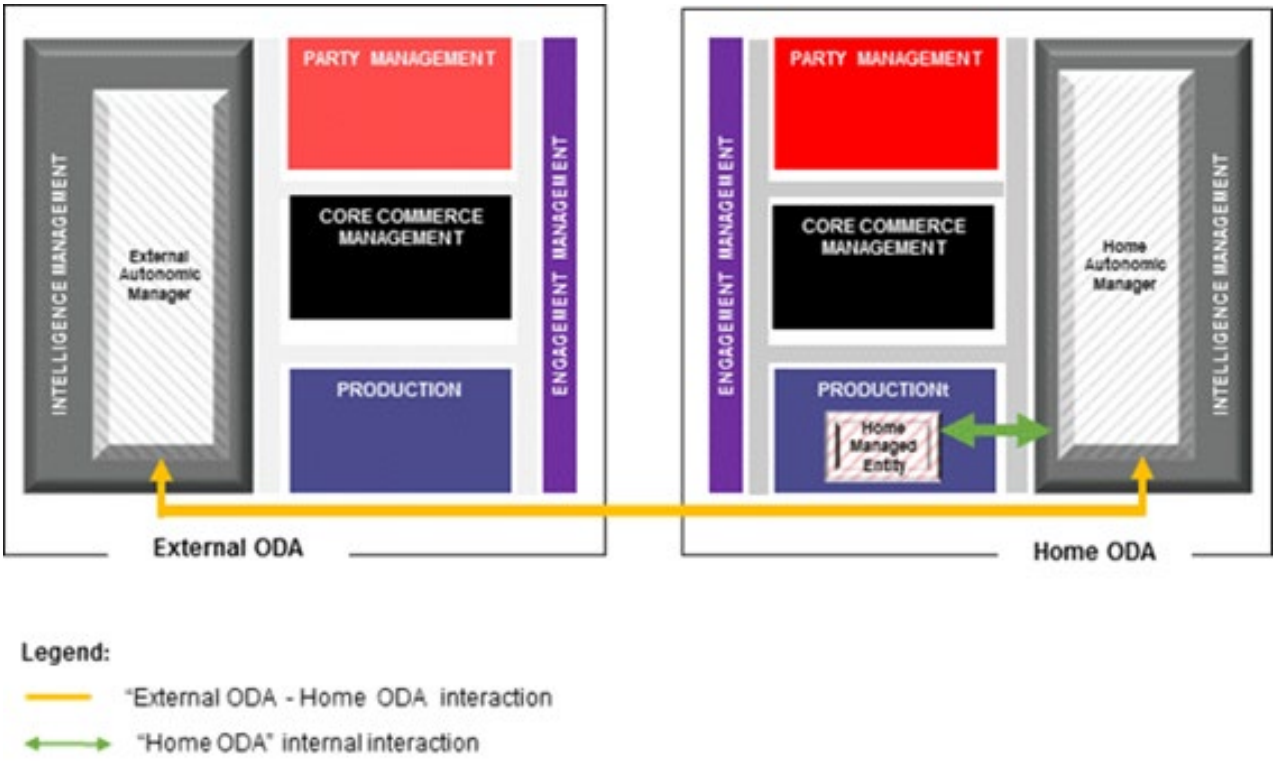

Figure 10.2. Federation of two ODAs (Peering / Sibling model)

Use Case 3: An Autonomic Manager belonging to an ODA is allowed to interact directly with a Managed Entity that is not its own

This use case illustrates the situation where two ODAs are collaborating from decision-making perspective when an External ODA needs to consume services exposed by some Home Managed Entities by policy-controlling them directly without the need to rely on the Home Autonomic Manager. There are some real use cases where such a service consumption model is relevant. This use case respects and enforces the Ownership rules because External Autonomic Manager cannot change the configuration of the Home Managed Entities (it is not allowed to do so).

Figure 10-3 depicts this use case. It is made of two "ODAs".

1. External ODA. It contains an "External Autonomic Manager". Its policy-controls the targeted Home Managed Entity. It is not allowed to change the configuration of the Home Managed Entity.
2. Home ODA. It contains a " Home Autonomic Manager" and the Home Managed Entity exposing services. Home Autonomic Manager has the right to change the configuration of this Home Managed Entity.
3. This interaction is shown by the yellow arrow.

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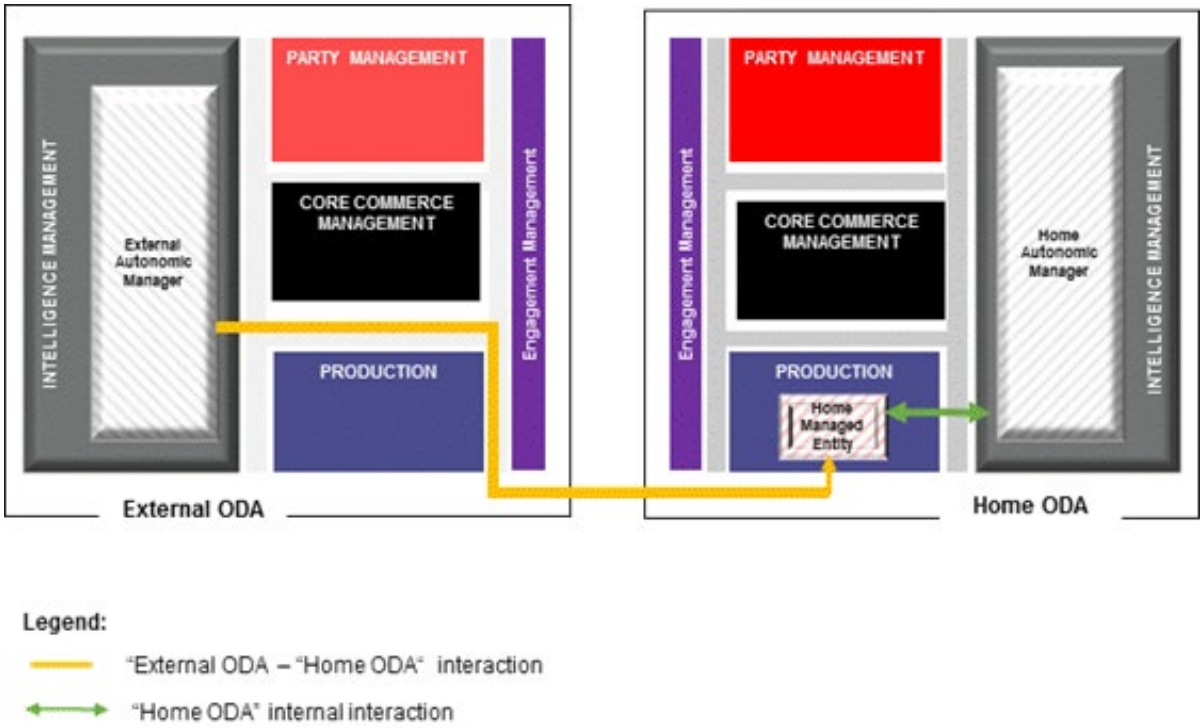

Figure 10.3. External Autonomic Manager interacting directly with Home Managed Entity that it does not own

Use case 4: An Autonomic Manager belonging to an ODA needs to communicate with a Managed Entity that is not its own via an Autonomic Manager that owns and manages this Managed Entity

This use case is a variant of use case 3. The only difference is that External Autonomic Manager is not allowed to interact directly with Home Managed Entities. For consuming service exposed by Home Managed Entities, External Autonomic Manager has to rely on the Home Autonomic Manager.

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This interaction is shown by the yellow arrow

Figure 9-4 depicts this use case

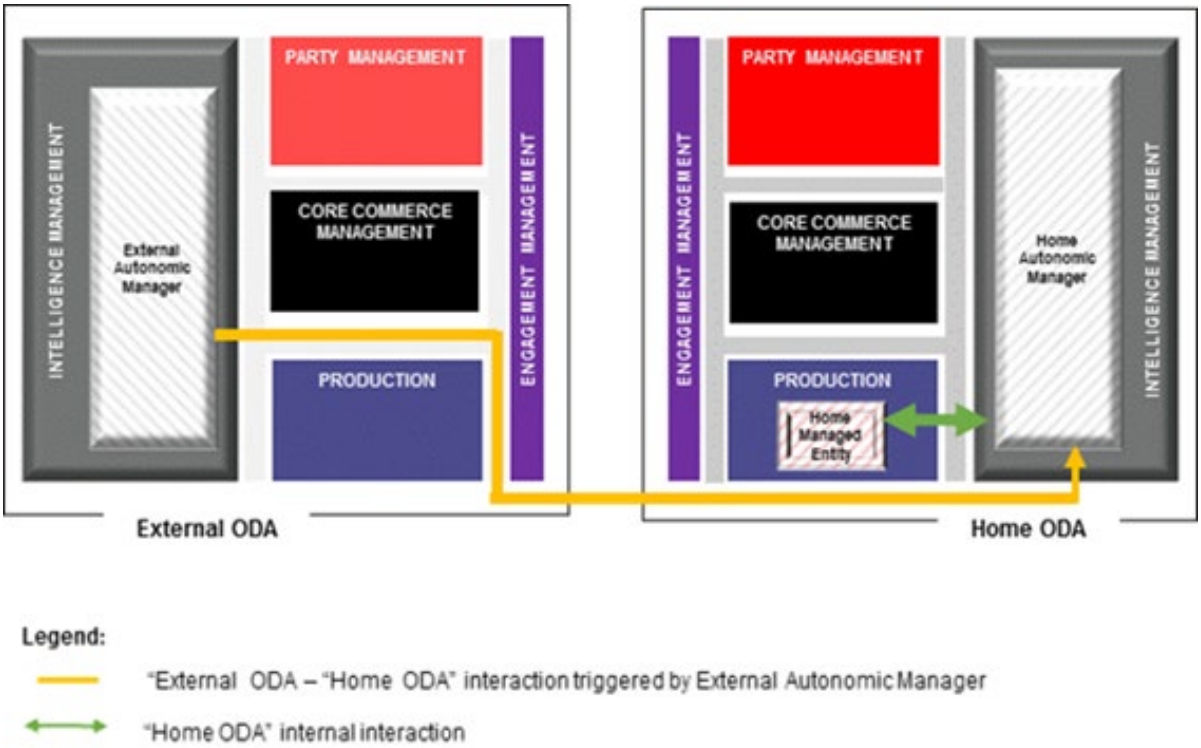

Figure 10.4. External Autonomic Manager communicates with non-owned Home Managed Entities via owning Home Autonomic Manager

10.1.2. Implementation flavors of ODA Knowledge Plane

ODA Knowledge Plane

The scope of the ODA Knowledge Plane is considered to be equivalent to that described in ETSI GANA (Generic Autonomic Network Architecture) Knowledge Plane specified in ETSI TS 103 195-2. As explained and depicted in Figure 9-5 (left-hand side) and in Figure 9-6, ODA Knowledge Plane is placed in the Intelligence Management Function Block. It is the Brain for Implementers to design and Implement Advanced Cognitive Management & Control DE Algorithms (not subject to standardization) to allow them to differentiate each other and make those Algorithms available to Digital Service Providers to select / choose / purchase the Best-in Class solutions.

This ODA Knowledge Plane is made of the following entities:

- Data Lake / Centralized / Shared Knowledge Base (cKB) populated by data from all the ODA Function Block and other sources (e.g., Events and KPIs) from PNFs/VNFs, Probes, Data Collectors. Raw Data collected are processed and translated onto Knowledge with common semantic and made available under Publish / Subscribe, Query & Find paradigm to be consumed by the centralized Decision-Making-Elements (DMEs) or Decision Elements (DEs).

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- Three categories of Knowledge Planes compose ODA Knowledge Plane:
    1. PM_KP: Party Management Knowledge Plane. It contains its Decision Elements (PM_DEs) with complex Analytics for Autonomics / AI (Cognitive) Algorithms
    2. CC_KP: Core Commerce Management Knowledge Plane. It contains its Decision Elements (CC_DEs) with complex Analytics for Autonomics / AI (Cognitive) Algorithms.
    3. P_KP: Production Knowledge Plane. It contains Decision Elements (CC_DEs) with complex Analytics for Autonomics / AI (Cognitive) Algorithm.

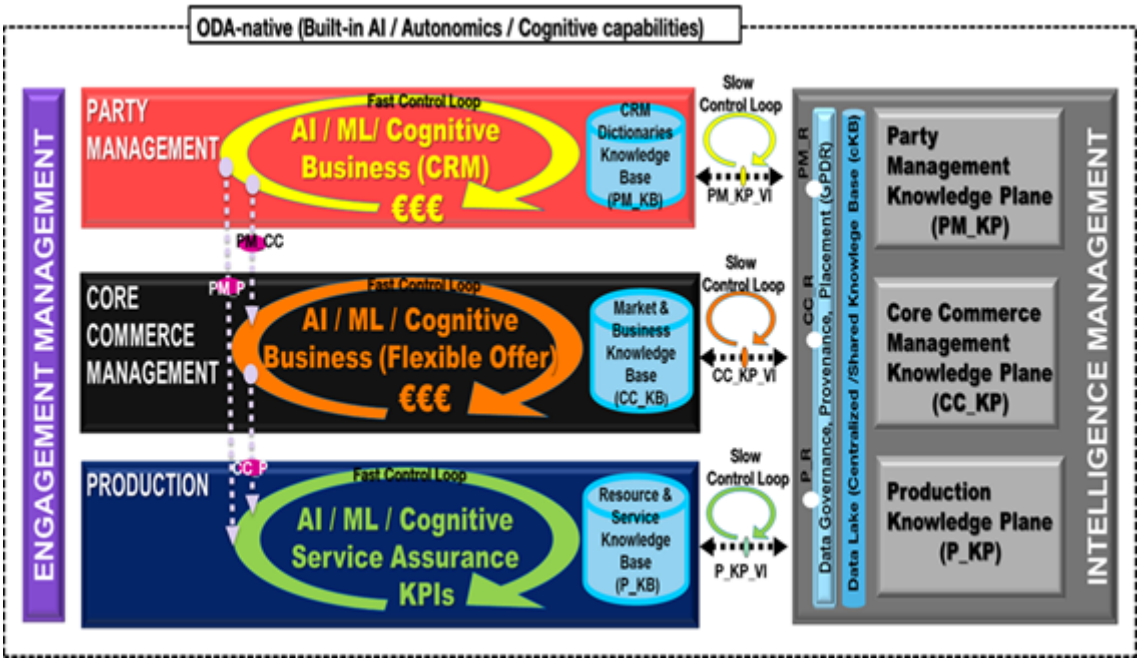

Figure 10.5. ODA Deployment option 1 (ODA-native)

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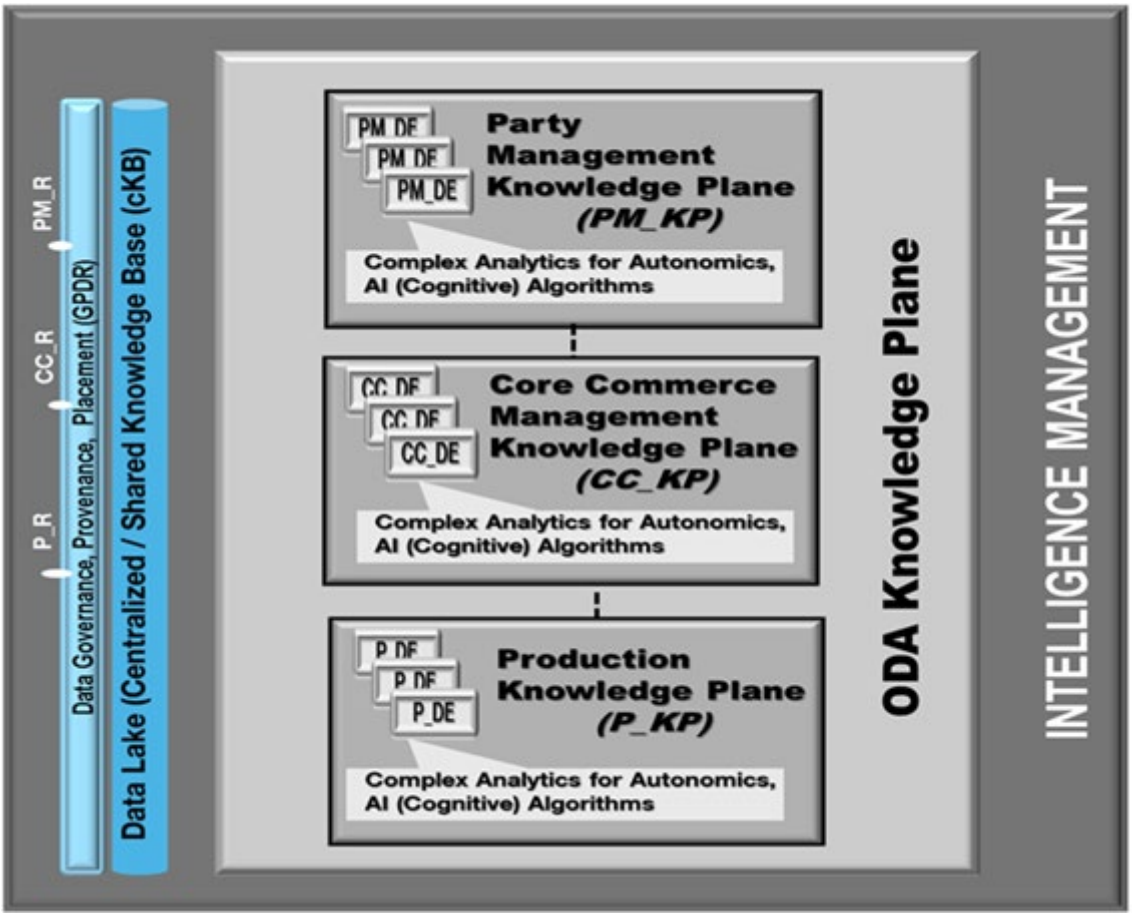

Figure 10.6. ODA Knowledge Plane

ODA Knowledge Plane interworking models

Figure A-7 depicts two models for the articulation and interworking between the three ODA Knowledge Planes:

- **Hierarchical model**

In this model, there is a need for building a top-level Knowledge Plane we named "ODA Master Knowledge Plane". This ODA Master Knowledge Plane is setting on the top and policy-controls the three other Knowledge Planes. To achieve this model, there is a need for defining reference points / Interfaces to exchange information / KPIs between ODA Master Knowledge Plane and the three ODA Knowledge Planes to perform a holistic decision-making across the three Knowledge Planes.

- **Federated model**

In this model, the three ODA Knowledge Planes are at the same level. The federation needs defining "East-West" Reference points / Interfaces between those ODA Knowledge Planes.

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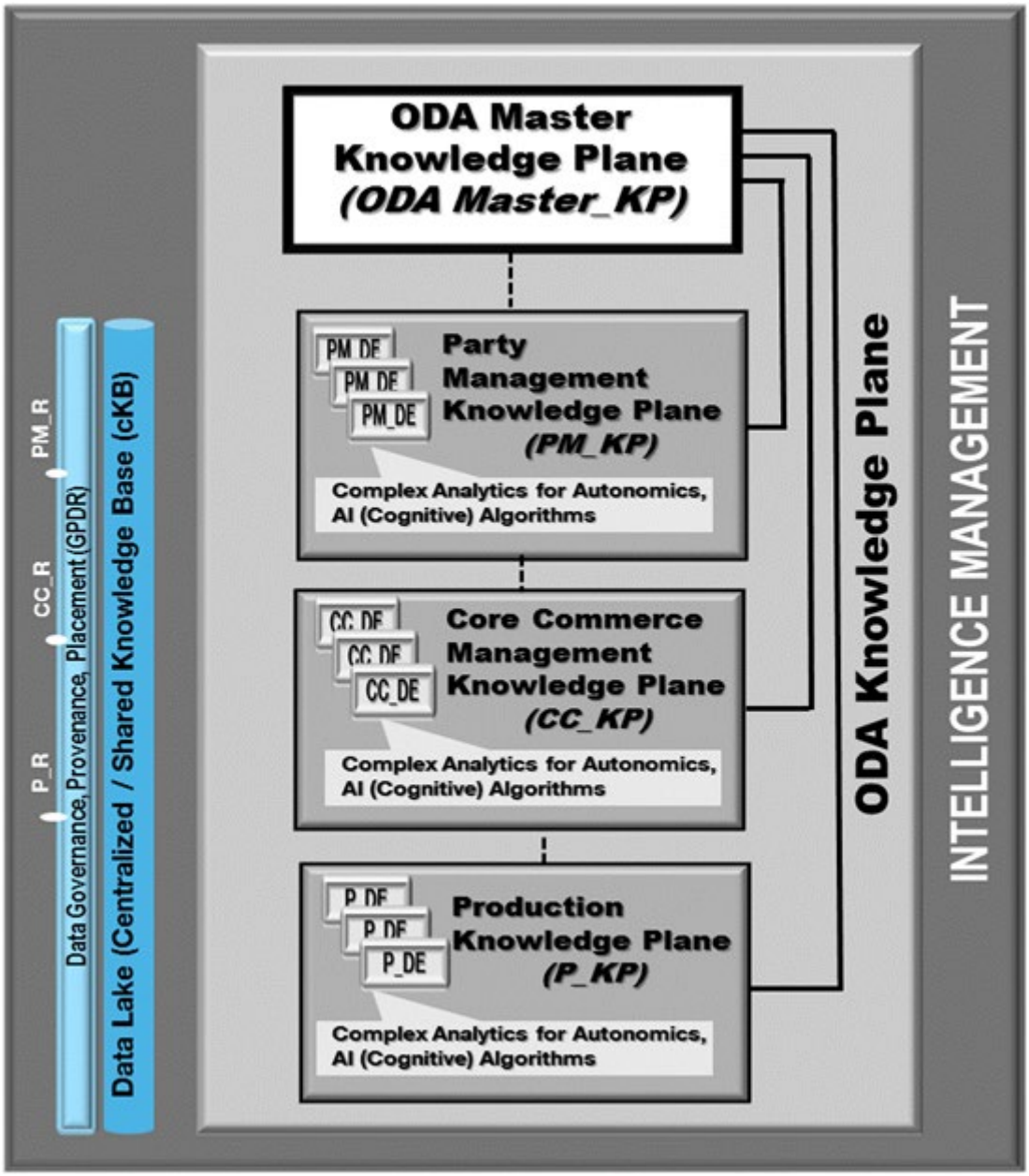

Figure 10.7. ODA Master Knowledge Plane (Hierarchy of Knowledge Planes)

- **Unique Knowledge Plane model**

It is to be noted that ETSI GANA TS 103194 specified a "unique Knowledge Plane" so called "GANA Knowledge / Dissemination Plane" who interacts with three layers (Business, Service, Infrastructure), http://www.etsi.org/deliver/etsi_ts/103100_103199/103194/01.01.01_60/ts_103194v010101p.pdf

Recommendation

ODA Functional Architecture Intelligence Management proposes these ODA Knowledge Plane models but does not prescribe any implementation solution. This is left to the Knowledge Planes and Algorithms designers / Implementers.

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Componentization view of Production Knowledge Plane (P_KP) (Centralized / Slow Control-loops) Autonomic Managers (DEs) and distributed /Fast Control-loops (DEs)

The concept of Knowledge Plane was introduced in ODA deployment option1 from design perspective and from a holistic view at each of the three horizontal ODA Function Blocks. The focus in this deployment option 2 is on componentization related aspect of the Decision-Making-Element (DME) or Autonomic Manager and its interfaces and associated primitives within the scope of Production Function Block.

The componentization of the ODA Autonomic Manager Elements (DMEs) either Knowledge Plane DMEs or distributed DMEs embedded in ODA Nodes is considered to be equivalent to that described in ETSI Generic Autonomic Network Architecture (GANA) Knowledge Plane specified in ETSI TS 103 195-2

- *ODA Autonomic Manager (DME) component view (Macro / Centralized -control loop or micro / distributed control loop)*

An Autonomic Manager Element (DME or DE) as a component is equipped with 3 categories of interfaces:

1. S_I: Sensory interface (the same interface is also embedded in the Managed Entity to allow the Autonomic Manager Element to policy-control Managed Entities)
2. Ef_I: Effector interface (the same interface is also embedded in the Managed Entity to allow the Autonomic Manager Element to policy-control Managed Entities)
3. DE to DE_I: Autonomic Manager Element to Autonomic Manager Element interface to allow an Autonomic Manager Element to federate with another one

- *ODA Node (embedding Distributed / Fast Control-loops or micro -Autonomic Manager Elements) component view*

This ODA Node is the counterpart of a Network Element (NE) in the Non-ODA platform, but it is made of Managed Entities (MEs) embedding micro Autonomic Managers Element(s). The ODA Managed Entity (ME) as a component is equipped with 4 categories of interfaces:

1. Ss_I: Sensory interface
2. Ef_I: Effector interface
3. SR_I: Service Requesting interface (it allows this Managed Entity to consume services provided by others)
4. SP_I: Service Providing interface (it is about the purpose of the Managed Entity it is designed for e.g., the service it provides)

Implementation flavors of ODA Autonomic Managers (DEs) as components

- ODA by design will follow principle of Cloud-Native which includes microservice -style implementation.
- ODA DMEs (DEs) will follow ETSI GANA implementation guide (TS 103 295-2) which considers DE as "overlay software" inserted to run on top of the resources layer (ODA Production Function Block) and recommends the two implementation flavors described below which are compliant with Cloud native and microservice principles:
- *Implementation flavor 1* - Individual Autonomic Managers (DEs) can be implemented as a singly merged run-time entity (e.g., a process)
- *Implementation flavor 2* - Individual Autonomic Managers (DEs) can be implemented to run as standalone processes (or threads for example) at run-time.

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How to introduce AI / Autonomic / Cognitive capabilities in ODA Production Function Block: case study

While the deployment option 2 depicted in Figure 10-9 is already covering Autonomic and Cognitive related aspects within Production Function Block, the main focus is at the fine grain of decomposition in the Production Function Block from an autonomic perspective. Autonomic Managers Centralized DME and Distributed DMEs) as well as Managed Entities (MEs) are considered as components, all interacting via their sensory interfaces and Effector interfaces and associated Operations / Primitives. In contrast, in the case study described here and depicted in Figure 10-8, we are looking at Autonomic and Cognitive capabilities in Production Function Block at "systems" level abstracting component view. In this regard, we consider Production Function Block following SDN / NFV deployment. The exercise consists in empowering this Production Function Block with Autonomic and Cognitive capabilities to make it "Autonomic & Cognitive-Aware". As described in Figures 10-5 and 10-10 we will consider the two levels of Autonomic Manager Elements (DMEs or DEs), centralized one and distributed one.

Step 1

- Embedding distributed DEs in the ODA Managed Entities of the Production Function Block, either Physical ones, we named "ODA PhyME" (ODA Production Physical Managed Entity) or virtualized ones we named "ODA_vME" (ODA Production virtualized Managed Entity),

Step 2

- Empowering "ODA Managers": ODA Service Orchestrator, ODA Network Orchestrator, ODA VNF Manager, ODA VIM, ODA SDN Controller (SDN-C), EMS by Production Knowledge Plane Decision Elements (P_DEs) logic. In this model, P_DEs software logic is considered as an application interacting with each ODA Manager via their Northbound interface / API.

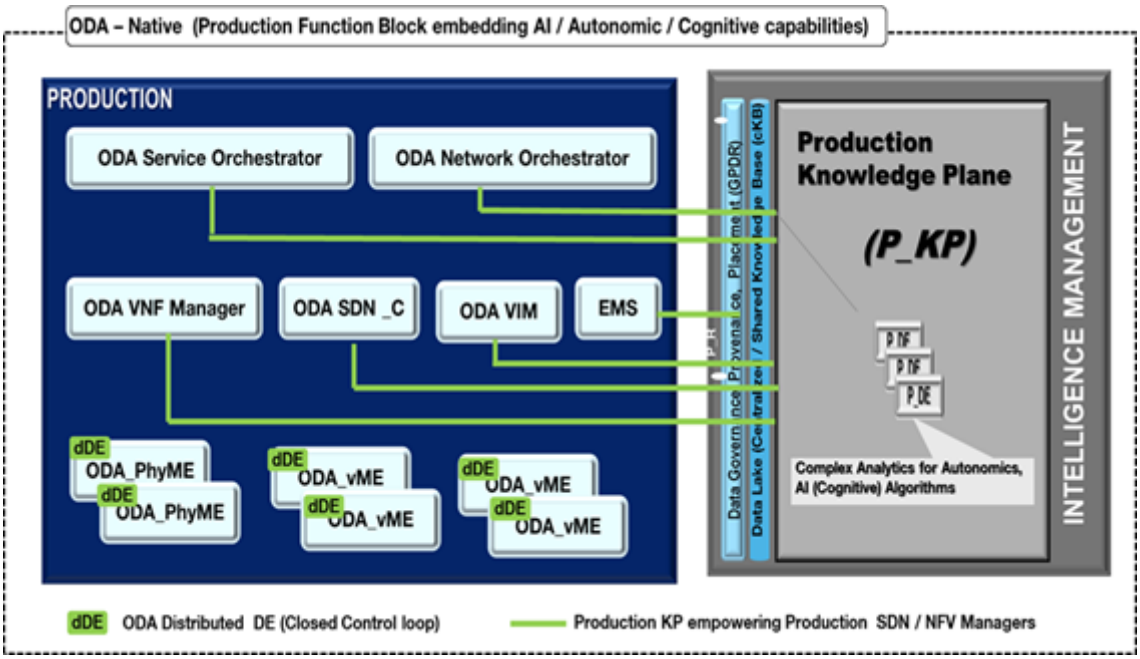

Figure 10.8. How to introduce AI / Autonomic / Cognitive capabilities in ODA Production Function Block: case study

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Relevance of this case study to Service Providers

This case study is of high interest to Service Providers who are facing various paradigms SDN / NFV, Service Orchestration, Big Data Analytics, Autonomic, Self- Management & Cognitive networking. Those paradigms all are sharing a common roadmap and targeting the same objective. Therefore, combining them constitutes a key enabler to 5G Network Slice Service Assurance. Extensive work in this space has been carried out by ETSI GANA, in one hand (see Note 1 below) and by a Multi-SDO joint activity, in the other hand (See Note 2 below)

Note 1: This use case is a high level adaptation from ETSI GANA case study (instantiation of ETSI GANA onto SDN, NFV, Service Orchestration) depicted in Figure 9-6 of http://www.etsi.org/images/files/ETSIWhitePapers/etsi_wp16_gana_Ed1_20161011.pdf

Note 2: This detailed architecture can be found in the Workshop Report from the Initiative: Joint SDOs/Fora Industry Harmonization for Unified Standards on AMC, SDN, NFV, E2E Orchestration, Software-oriented enablers for 5G: Joint SDOs/Fora Workshop Report from the workshop held on 4th June 2015 at the TM Forum Live 2015 http://projects.sigma-orionis.com/eciao/wp-content/uploads/2015/07/Report-on-Joint-SDOs-Industry-Harmonization-for-Unified-Standards-on-AMC_SDN_NFV_E2E-Orchestration_ver3.01.compressed.pdf

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11. BUSINESS SCENARIOS / USE CASE

11.1.1. ODA and legacy BSS/OSS (INFORMATIVE)

Legacy to ODA-native deployment option in a Transition phase

Figure 11-1 depicts this deployment option. In this case we consider the ODA Intelligence Management (Autonomic / AI / Cognitive) capabilities are embedded in the Production Function Block only which is the Function Block of the ODA Functional Architecture that is relevant from migration path perspective when it comes to introduce ODA at an early stage, it shows the interaction with a Non-ODA Platform. The main characteristics and requirements associated with this deployment option are:

- From the ODA Knowledge Plane perspective, as depicted in Figures 9-5 and 9-6, we consider only P_KP (Knowledge Plane of the Production Function Block). P_KP shall interwork with Legacy OSS. To this end, there is a need for specifying:
  - a reference point / API between the Legacy OSS and the Production Knowledge Plane (P_KP) we named “OSS_(P_KP)
  - a reference point / API between the Non-ODA Nodes (Legacy Managed Entities, in this case it is about Managed Elements in traditional management-style sense) and the Production Knowledge Plane (P_KP). We named this reference point “L_(P_KP)” (Legacy to P_Knowledge Plane)
- The Production Knowledge Plane (P_KP) shall embed capabilities to manage Non-ODA Managed Entities (MEs) or Network Elements (NEs) for the purpose of the transition phase. This is the reason why there is a need for specifying this “L_(P_KP)” reference point.
- There is no need to evolve (upgrade) the Legacy OSS not only for cost saving reasons (upgrading the legacy OSS is time and cost consuming) but mainly for moving away from the Legacy OSS to the ODA-native-style which embeds advanced Management & Control capabilities (it is “Intelligent and Cognitive i.e., Autonomic per design”). Those Autonomic & Cognitive capabilities are enablers for new value stream creation mainly in the 5G Network Slice space for verticals.
- Well-Known traditional management interfaces associated with Non-ODA Platforms are indicated:
  - SBI (South Bound Interface: between Legacy Managed Entities (MEs) and EMS (Element Management System)
  - NBI (North Bound Interface) between EMS and NMS (Network Management System)

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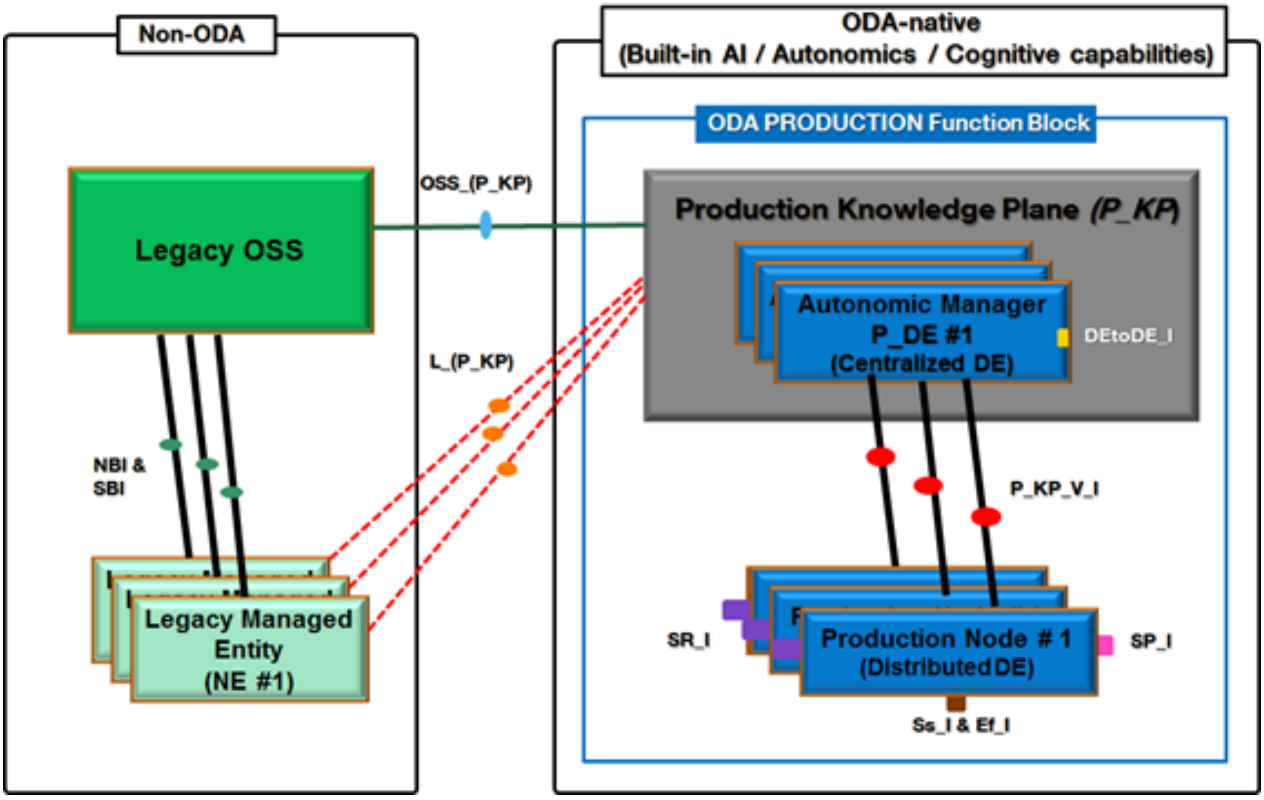

Figure 11.1. ODA Deployment option 2: Legacy to ODA

11.1.2. Managing Federation of ODA

ODA Intelligence Management Federated Framework

a) Federation under Multiple ODA Knowledge Planes scenario

Figure 11-2 depicts this deployment option 3. In this case, two ODA-native instances (ODA #1 and ODA #2) are interworking within a federated model. The use case behind this federated model is the case where ODA #1 belongs to a Digital Service Provider DSP #1 and ODA #2 belongs to a Digital Service Provider DSP #2. Each ODA instance is embedding AI / Autonomic / Cognitive capabilities at the three horizontal Function Blocks of the ODA Functional Architecture as depicted in Figure 10-5 (ODA-native deployment option 1).

The main characteristics and requirements associated with this option are:

- The *Party Management Closed Control Loop* of ODA #1 needs to exchange information, KPIs with the *Party Management Closed Control Loop* of ODA #2 for E2E Cross –ODA decision-making. There is a need for specifying a reference point / API between those two Closed Control Loops we named “PM_PM". This federation is at Knowledge Panes levels (Party Management Knowledge Plane) "PM_KP" as defined in deployment option 1.
- The *Core Commerce Management Closed Control Loop* of ODA #1 needs to exchange information, KPIs with the *Core Commerce Management Closed Control Loop* of ODA #2 for E2E Cross-ODA decision-making. There is a need for specifying a reference point / API between those two Closed Control Loops we named “CC_CC”. This federation is at Knowledge Panes levels (Core Commerce Management Knowledge Plane) "CC_KP" as defined in deployment option 1.

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- The *Production Closed Control Loop* of ODA #1 needs to exchange information, KPIs with the *Production Closed Control Loop* of ODA #2 for E2E Cross-ODA decision-making. There is a need for specifying a reference point / API between those two Closed Control Loops we named "P_P". This federation is at Knowledge Plane levels (Production Knowledge Plane) "P_KP" as defined in deployment option 1, It corresponds to the interworking between the two Knowledge Planes *as specified by GANA (ETSI TS 103 195-2).*

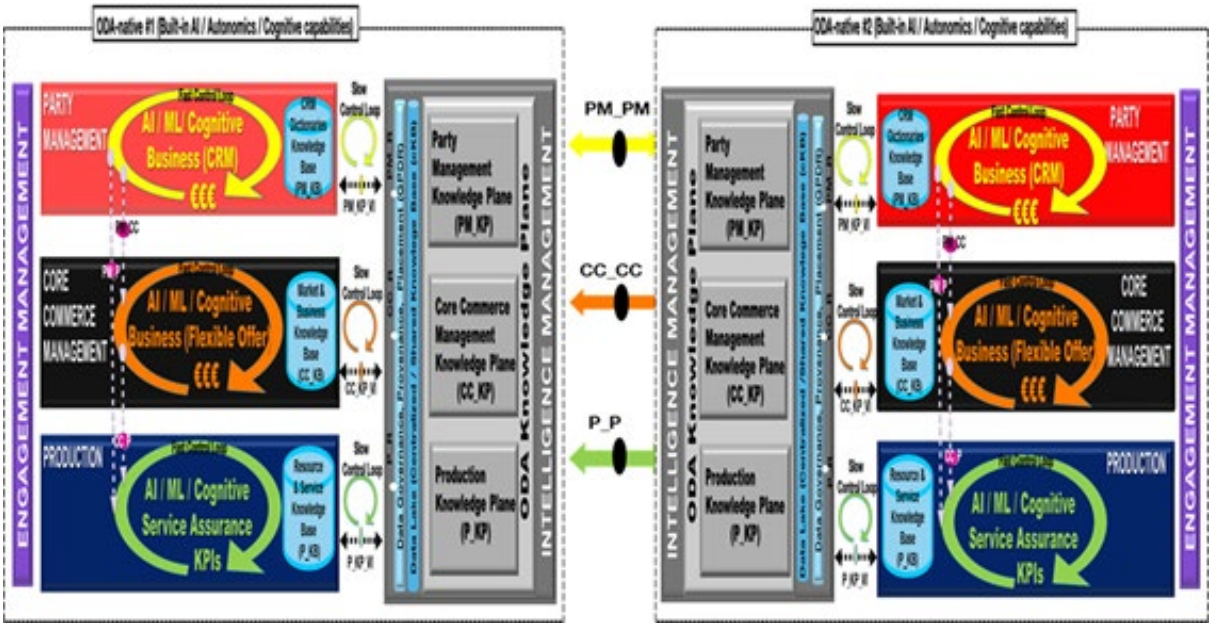

Figure 11.2. ODA Intelligence Management Federated Framework (multiple ODA Knowledge Planes scenario)

a) Federation under a single ODA Knowledge Plane scenario

Figure 11-3 depicts the same Framework as the one depicted in Figure 11-2, but ODA Intelligence Management is composed of a single (Unique) Knowledge Plane that can interact with the four other ODA Function Blocks.

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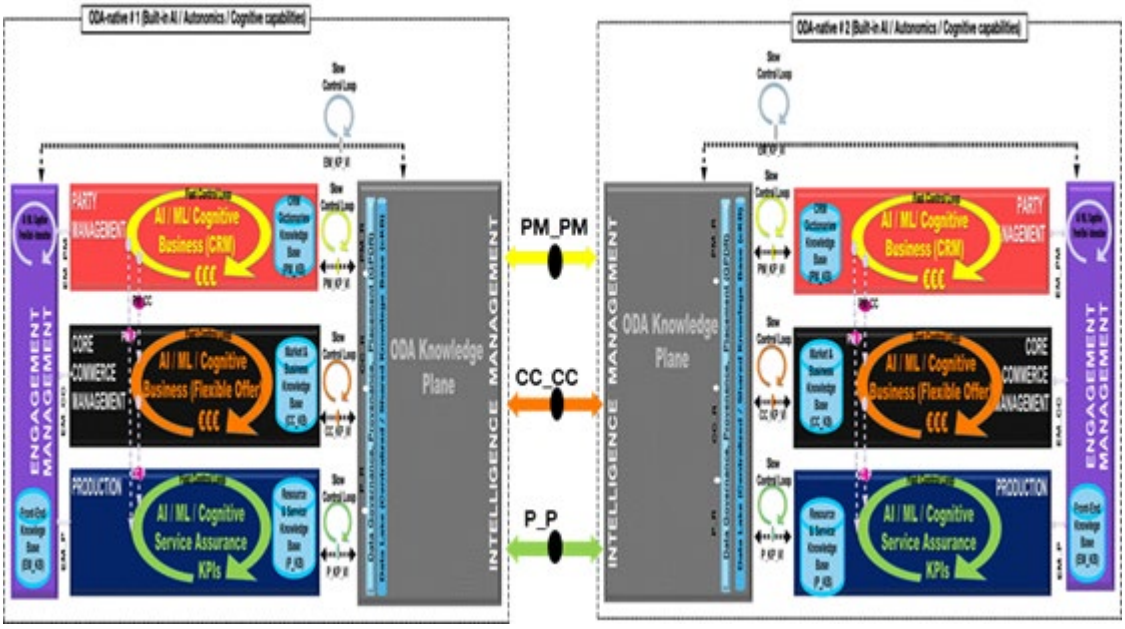

Figure 11.3. ODA Intelligence Management Federated Framework (Single / Unique ODA Knowledge Plane scenario)

Note - This Federated Framework focuses on Federation of two ODA-native instances, but we think this federation model could apply to interworking / federating between an ODA-native instance and any other "System" / "Platform" (e.g., ONAP Platform, Vendor-specific Platform) as far as this peer Entity exhibits / exposes the interworking reference points / interfaces described earlier.

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